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Technical Report No. 13 (Revised 2022)

Fundamentals of an Environmental Monitoring Program

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Fundamentals of an Environmental Monitoring Program

Technical Report No. 13 (Revised 2022)

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1.0 Introduction

The design and implementation of an environmental monitoring (EM) program allows the confirmation that controls to minimize the risk associated with processes, people, equipment, utilities, and facilities are designed correctly and continue to be effective in the manufacture of sterile pharmaceutical products. As such, it forms an important part of the contamination control strategy (CCS).

PDA first published *Technical Report No. 13* in 1990 and revised it two previous times, 2001 and 2014. This third revision intends to align the content of the document with current industry trends and regulatory expectations. Specifically, it will provide additional guidance and focus on the:

- Use of quality risk management (QRM) principles to establish a robust EM program by suggesting practical step-by-step approaches
- Increased expectations concerning data management and data integrity
- Prerequisites to an effective EM program
- Qualification of controlled environments and how to maintain them in a qualified state
- Rapid microbiological methods (RMM) or automated monitoring systems, since becoming more and more convenient, performant and accepted, or even recommended, by regulatory bodies.

Members of the task force who developed this technical report represent or provide expertise to global companies to ensure that the methods, terminology, and practices reflect not only U.S., but also international procedures. Technical peer reviews were completed by prominent EM scientists.

Although this TR has been updated to ensure its alignment with existing current regulations and guidelines, current and/or newly updated regulations or guidelines should be consulted when implementing or changing an environmental monitoring program. This TR should be viewed as technical guidance, it is not intended to replace existing or future regulations, nor should it be considered an all-inclusive document.

1.1 Purpose

This document was created to aid in the establishment of an environmental monitoring program that is robust, meaningful, and practical whilst embracing innovation and the principles of quality risk management. This revision updates microbiological and total airborne particulate control concepts and principles as they relate to facilities involved in the manufacture of sterile pharmaceutical products and other designated cleanroom environments.

1.2 Scope

TR-13 serves as a resource applicable for controlled environments in general however its primary focus is environmental monitoring for sterile product manufacturing.

This document discusses international standards and regulatory guidances, elements of an EM program, and EM application. Current (at the time of publication) guidelines for typical EM levels and frequencies are presented in **Section 3.0** and levels for pharmaceutical water are covered in **Section 6.8**.

A comprehensive environmental monitoring program should demonstrate the effectiveness of a solid CCS focusing on:

- Sound facility design, including barrier systems (e.g., isolators and RABS), operation and maintenance
- Established documentation systems
- Qualified sanitization, disinfection, and decontamination procedures
- Reliable process controls
- Good housekeeping practices
- Effective area access controls
- Consistent sample collection and analysis

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- Effective training, certification or qualification, and personnel evaluation programs (e.g., periodic shop/manufacturing-floor oversight programs by a quality assurance unit and/or adequate subject matter experts)
- Quality assurance and control of materials, facilities, and equipment
- Continuous improvement and quality risk management programs

Bulk and intermediate product bioburden monitoring (except for pharmaceutical water) is not considered part of all EM programs and is therefore outside of the scope of this technical report. For more specifics on this topic, refer to PDA *Technical Report No. 69: Bioburden and Biofilm Management in Pharmaceutical Manufacturing Operations* ***(1)***.

2.0 Glossary and Abbreviations

Action Limit
A microbial or total airborne particulate limit that, when exceeded, indicates a process is outside of its normal operating range. A response to such an excursion should involve a documented investigation and corrective actions based on the results of that investigation.

Airborne Particulate Count (Total Airborne Particulate Count)
The total number of particles of a specific size range per unit volume of air.

Airborne Viable Particulate Count (Total Airborne Aerobic Microbial Count)
The recovered number of colony-forming units or related signal per unit volume of air.

Alert Level
An established value that when exceeded, provides an early warning of potential drift from normal operating conditions and validated state, which does not necessarily give grounds for corrective action but triggers appropriate scrutiny and follow-up to address the potential problem.

Aseptic Filling
Part of aseptic processing in which a pre-sterilized or aseptically manufactured product is filled and sealed into sterile containers in a cleanroom, RABS, or isolator.

Aseptic Processing
Handling of product, packaging materials, and/or processing equipment in a controlled environment in which the air supply, materials, equipment, and personnel are controlled to maintain (product) sterility ***(2)***.

Bioburden
Viable microorganisms associated with personnel, manufacturing environments (air and surfaces), equipment, product packaging, raw materials (including water), in-process materials, or finished products.

Campaign
A series of consecutive production batches manufactured without or with reduced intervening cleaning, disinfection, and/or sterilization.

Cleaning
Chemical or physical means used to remove soil and/or microorganisms from surfaces.

Cleanroom
A room designed, maintained, and controlled to prevent particle and microbiological contamination of a drug product or medical device. A cleanroom is assigned and reproducibly meets an appropriate air cleanliness classification.

Colony-Forming Unit (CFU)
A single macroscopic colony formed after the growth of one or more microorganisms on a solid microbiological growth medium.

Contamination Control Strategy (CCS)
A holistic, risk-based, planned, and documented approach based on product and process understanding, which explains all the elements that need to be controlled and monitored so micro-

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bial, particulate, and pyrogen contamination is minimized to ensure process performance and product quality. The CCS should allow for continuous improvement measures to be taken.

Continuous Monitoring
A process of data collection in which conditions are monitored continuously throughout the operation. In most applications, this definition implies "during production."

Controlled Non-Classified Area (CNC)
An area that does not meet any of the formal classified grades of cleanliness (i.e., grades A to D), but where a manufacturer defined level of microbial control is still maintained. (May also be referred to as Pharmaceutically Controlled Clean Area (PCCA))

Corrective Action
Action to eliminate the cause of a nonconformity and to prevent recurrence.

Critical Area/Critical Zone
An area designed to maintain sterility of materials where sterilized product, containers, closures, and equipment may be exposed to the manufacturing environment.

Critical Surface
A surface within a critical area that may come in direct or indirect contact with sterilized or sterile filtered products, containers, or closures.

Decontamination
The overall process of removal or reduction of any contaminants (chemical residue, or microorganisms) from an area, processing equipment, or person. The method of decontamination used (e.g., cleaning, disinfection, sterilization) should be chosen and validated to achieve a level of cleanliness appropriate to the intended use of the item decontaminated.

Disinfection
The chemical or physical inactivation of a bioburden on inanimate surfaces ***(2)***.

Grade A Air Supply
Air which is passed through a filter qualified as capable of producing Grade A total particulate quality air, but where there is no requirement to perform continuous total particulate monitoring or meet Grade A viable monitoring limits and the area itself is not classified. Specifically used for the protection of fully stoppered vials where the cap has not been crimped and the equipment and engineering systems that have a direct impact on product quality.

Grid Profiling
A process of dividing areas of equivalent classifications into grids for the purpose of uniformly assessing contamination characteristics in that area.

Isolator
A contained, decontaminated environment meeting Grade A (ISO 5) conditions used for aseptic process manufacturing that provides an uncompromised, continuous isolation of its interior from the external environment including personnel. Once decontaminated by a validated cycle, an isolator prevents the microbiological contamination of sterile products and product contact surfaces of the interior by enclosures and the supply of continuous, controlled overpressure of HEPA-filtered air.

Closed Isolator
Systems exclude external contamination from the isolator's interior by transfer material via aseptic connection to auxiliary equipment, rather than using openings to the surrounding environment. Closed systems remain sealed throughout operations.

Open Isolator
Systems are designed to allow for the continuous or semi-continuous ingress and/or egress of materials during operations through one or more openings. Openings are engineered (e.g., using continuous overpressure) to exclude the entry of external contamination into the isolator.

Manufacturing Control Parameters
Conditions and corresponding measurements associated with facilities and equipment used in the control of a manufacturing process and/or area that may impact the safety, quality, identity, potency, or purity of a product. Environmental control parameters may include airflow rates and patterns, pressure differentials, materials and personnel flow, temperature, and relative humidity, as well as total airborne total particulate and viable particulates. Process and operating control parameters include bioburden, process rate, weight, volume, temperature, and pressure.

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Microbial Characterization
The description of microorganisms based on their cellular morphology, Gram reaction, and key diagnostic tests (e.g., gram-positive coagulase-negative cocci).

Microbial Classification
The arrangement of microorganisms into taxonomic groups based on their similarities and relationships.

Microbial Identification
The determination of the genus, and species, when possible, to which a laboratory or manufacturing isolate belongs.

Operational State
The status of a classified room often used an as indicator of the presence or absence of personnel and/or active manufacturing processes.

At Rest
Condition of the environment in the absence of normal operations. This includes having the equipment installed as specified and is standing by for operation when no personnel are present. (Synonymous with "static.")

In Operation
Condition of an environment during normal operations, that is, when the usual equipment is operating and the specified number of personnel are present, and the process or simulated process is ongoing. (Synonymous with "dynamic.")

Parametric Release
A sterility release program based on effective control, monitoring, and documentation of a validated sterile-product manufacturing process where sterility release is based on demonstrated achievement of critical operational parameters in lieu of end-product sterility testing *(3)*.

Preventive action
Action to eliminate the cause of a potential nonconformity or other potential undesirable situation.

Restricted Access Barrier System (RABS)
Aseptic processing systems (typically ISO 5) intended to substantially reduce human-borne contamination within the aseptic environment where sterile product, containers, closures, and equipment can be exposed via the use of separative devices and defined mechanical features and operating procedures *(4)*.

Open RABS
Where there are vents in the barrier that allow air to move from the RABS to the surrounding area (for example, Grade A zone to the surrounding Grade B area).

Closed RABS
Where the air is vented in return ducts within the cabinet.

Risk Analysis
The estimation of the risk associated with the identified hazards *(5,6)*.

Sanitization
Process of reducing the number of all forms of microbial contamination including fungi, viruses, and bacteria *(7)*.

Spore
A bacterial dormant form that is highly resistant to adverse conditions. Fungal spores are not highly resistant; their susceptibilities are closer to vegetative microorganisms.

Sterilization
Validated process used to render product, equipment, or primary packaging materials free from viable microorganisms *(7)*.

Type Strain
A specific isolate of a species that is maintained in pure culture and is serotypically, genotypically, or chemotaxonomically characterized to differentiate it from other strains of the same species. The strain is representative of the species and provides a reference for the species based on its historic isolation, characterization, and deposition in recognized culture collections.

Terminal Sterilization
A process where the material is sterilized in its primary packaging/container configuration.

Total Airborne Particulates
A term used in reference to the total number of particulates suspended in the air. Airborne particulates may be a complex mixture of viable and nonviable particulates. They can be characterized by their physical attributes (e.g., concentration and size distribution).

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Trend Analysis
A review performed in response to an alert or action condition or as part of periodic data review. This review provides an analysis of specific environmental monitoring data to identify adverse trends (e.g., increase in counts, a change to repetitive organism recoveries, or multiple counts in the same room).

Vegetative Cell
Cells in an actively growing state. Some microorganisms can only be vegetative, while others are spore-formers and can be in a vegetative or spore (dormant) state.

2.1 Abbreviations

API Active pharmaceutical ingredient

CCS Contamination control strategy

CRR Contamination recovery rate

EM Environmental monitoring

FMEA Failure Mode & Effects Analysis

HACCP Hazard Analysis Critical Control Point

HEPA Filter High-efficiency particulate air filter

RABS Restricted access barrier system

SCDA Soybean casein digest agar (synonymous with TSA)

SDA Sabouraud dextrose agar

TSA Trypticase soy agar or tryptone soya agar (synonymous with SCDA)

VBNC Viable but not culturable

3.0 Environmental Classifications: Regulatory Expectations

An EM program should be designed and implemented in conformance with the requirements of the government agencies regulating the manufacturing site as well as requirements of international standard-setting organizations. If the intent is to serve multinational markets, the most stringent requirements should be evaluated as the basis of an EM program.

This section compares regulations and guidance instituted for EM by these authorities and standards-setting organizations:

- International Organization for Standardization (ISO)
- U.S. Food and Drug Administration (FDA)
- U.S. Pharmacopeia (USP)
- European Pharmacopoeia (Ph. Eur.)
- European Medicines Agency (EMA)
- Ministry of Health, Labour and Welfare of Japan (MHLW)
- Japanese Pharmacopoeia (JP)
- Chinese Pharmacopeia (ChP)
- Pharmaceutical Inspection Convention (PIC) and Pharmaceutical Inspection Co-operation Scheme (PIC/S)
- World Health Organization (WHO)

Although the regulations and guidelines of these organizations are similar in many respects, there are important differences among them in terms of the information each provides, particularly with respect to cleanroom classifications. Specific countries other than those listed above may have additional regulations and guidances.

The most commonly accepted international cleanroom standard related to total airborne particulates is *ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of Air Cleanliness by Particle Concentration*. ISO class designations are based on the number of particles greater than a specified size (0.1–5 µm) per cubic meter of air sampled. ISO 14644-1 defines classes from 1 to

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9, with ISO 1 being the cleanest. ISO classes 5 through 8 are used in the pharmaceutical industry for sterile-product manufacture and other areas where airborne particulate control is required ***(8)***.

Some international and national bodies have based their cleanroom requirements for sterile manufacturing on the ISO standard, thus it is used here as the benchmark for other regulations and standards. USP has adopted the ISO cleanliness classes in its *General Guidance Chapter <1116> Microbiological Control and Monitoring of Aseptic Processing Environments* ***(9)***.

Some international regulations, for example, those from the People's Republic of China, require ISO-class cleanrooms for the manufacture of all types of nonsterile products. China's National Medical Products Administration requires ISO 8 tested at rest for areas where nonsterile product is exposed during manufacture. The EU good manufacturing practice (GMP) regulations, however, require ISO 8 classification only for the manufacture of inhalants. In other jurisdictions, firms are free to apply the ISO classes to nonsterile manufacturing or to specify equivalent or modified conditions based on their product requirements. In general, environmental control and monitoring in non-sterile product manufacturing is out-of-scope of this technical report. There are other resources available which cover this topic, including USP <1115> *Bioburden Control of Non-sterile Drug Substances and Products*.

ISO does not specify the operating state of the area being classified (i.e., as built, at rest, or in operation), nor does it specify the particle-size thresholds to be employed. In addition, the ISO standard deals only with total particulate count and does not discern between viable and nonviable particles. The regulating or advisory body determines those specifications. Therefore, each regulatory or standard-setting body has comparable, but somewhat different, applications of the ISO standard. The following are activities generally expected to take place in each of the ISO classifications:

- **ISO 5:** Aseptic processing zone; sterile product and/or packaging component exposed; unidirectional airflow required
- **ISO 7:** Area immediately surrounding the aseptic processing zone
- **ISO 8:** Non-sterile formulation, materials, and component preparation; filling area for terminally sterilized product

The EU, PIC/S members, and WHO use an alphabetic classification of Grades A–D ***(10-12)***. For each grade, a ≥0.5 µm and ≥5 µm particle-count limit is specified (see **Table 3.0-1**). In addition, these limits are set for both at-rest and in-operation states. Thus, each grade covers up to two corresponding ISO classes, as follows:

- **Grade A:** ISO 5 at rest and in operation (ISO 4.8 if based on particles ≥5 µm)
- **Grade B:** ISO 5 at rest, ISO 7 in operation
- **Grade C:** ISO 7 at rest, ISO 8 in operation
- **Grade D:** ISO 8 at rest, undefined in operation

Japan has adopted similar classifications in its *Guidance on the Manufacture of Sterile Pharmaceutical Products by Aseptic Processing.* One minor difference is that the Japan guidance refers to the Grade A classification as ISO 5 rather than ISO 4.8, although the limit for particles ≥5 µm technically corresponds to ISO 4.8. The *Japanese Pharmacopeia* also uses the Grade A–D classification system but does not specify counts ≥5.0 µm and refers to the U.S. FDA guidance for equivalency in operation ***(13)***.

For the United States, the FDA's 2004 *Guidance for Industry Sterile Drug Products Produced by Aseptic Processing: Current Good Manufacturing Practice* and USP <1116> both discuss the application and environmental requirements for cleanrooms used for aseptic processing. ***(9, 14)*** These include ISO classes 5–8 and their corresponding *Federal Standard 209E* classes (e.g., ISO 5/Class 100). Limits are set only for particles ≥0.5 µm. Both are silent on the state of the area to be tested but clearly imply areas to be "in operation."

Common factors among the various guidelines and regulations described here include the requirement that ISO 5 areas, where aseptic conditions must be maintained, typically require unidirectional airflow.

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ISO 14644-1 describes unidirectional flow as 0.45 meters per second (90 ft/min) plus or minus 20% measured 150–300 mm from the supply filter face. EU Annex 1, though, recommends this measurement be taken as close to the work surface as practical ***(10)***.

In addition, all regulatory authorities recommend an air pressurization scheme to ensure airflow from the cleaner zone to the less-clean zone. The broadly acceptable guidance value is a differential pressure (ΔP) of 10–15 Pa (0.04–0.06 in H_2O) between zones of differing classes. Several of the relevant guidance documents describe the use of air locks to maintain the differential pressures while doors are used between different Grades. Where an open ISO 5 unidirectional zone is placed within an ISO 7 background, this pressure differential is not required.

Table 3.0-1 summarizes the area maximum particulate limits for sterile manufacturing specified by various regulatory and standard-setting organizations. **Table 3.0-2** summarizes monitoring requirements for the various environments according to the relevant authorities. The summary of requirements is condensed and is limited to routine monitoring; it excludes requirements for area qualification or requalification. More complete guidance may be found in the references cited.

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Table 3.0-1 Cleanroom Monitoring – Airborne Particulate Limits (particles/m^3)

Particle Size	ISO 14644	U.S. FDA (Aseptic Processing Guidance) [1,2]	USP <1116>	EU & PIC/S Annex 1 and WHO Annex 6	Japan MHLW (Aseptic Processing Guidance)	JP XVI
	ISO 5	ISO 5/Class 100	ISO 5	Grade A/ Grade B (at rest)	Grade A/ Grade B (at rest)	Grade A/ Grade B (at rest)
≥0.5 µm	3,520	3,520	3,520	3,520	3,520	3,520
≥5.0 µm	Not specified	Not specified	Not specified	20[3] /29	20[3]	Not specified
	ISO 7	ISO 7/Class 10,000	ISO 7	Grade B (in operation) Grade C (at rest)	Grade B (in operation) Grade C (at rest)	Grade B (in operation) Grade C (at rest)
≥0.5 µm	352,000	352,000	352,000	352,000	352,000	352,000
≥5.0 µm	2,930	Not specified	Not specified	2,900	2,900	Not specified
	ISO 8	ISO 8/Class 100,000	ISO 8	Grade C (in operation) Grade D (at rest)[4]	Grade C (in operation) Grade D (at rest)	Grade C (in operation) Grade D (at rest)
≥0.5 µm	3,520,000	3,520,000	3,520,000	3,520,000	3,520,000	3,520,000
≥5.0 µm	29,300	Not specified	Not specified	29,000	29,000	Not specified

1 Obsolete U.S. Federal Standard 209E classification added for continuity

2 Class titles for U.S. FDA and USP indicate equivalent particle counts per cubic foot

3 ISO 4.8 is based on reduced limit for particles ≥5 µm. European regulation is considering removing the ≥5µm classification for Grade A and Grade B at-rest. However, as it is still an expectation within the current EU Annex 1 regulation and other standards, so this particle-size requirement remains within the table.

4 Grade D operational particulate counts depend on the operation and are not defined by any guideline.

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Table 3.0-2 Environmental Monitoring Requirements/Guidance[1]

Monitoring Guidance	U.S. FDA (Aseptic Processing Guidance)	USP <1116>[3]	EU Annex 1, PIC/S and WHO Annex 4 [2]	Japan MHLW (Aseptic Processing Guidance)	JP XVI
Frequency (Airborne total particulate and viable count, surface viable count, personnel sampling as noted)	**ISO 5:** Each production shift. Gloves/Gown each exit, daily or each lot. Other classes not specified.	Cleanroom/RABS Critical Zone (ISO 5 or better): • Each operational shift (active air) • Surface monitoring (end of operation) Aseptic area adjacent critical zone: • Each operating shift (all sampling) Other non-adjacent aseptic areas: • Once per day (all sampling) Isolators Critical Zone (ISO 5 or better): • Once per day (active air) • Surface monitoring (end of campaign) Non-aseptic areas surrounding the isolator: • Once per month (all sampling)	**A:** In operation, continuous particulate monitoring required for critical operations. Frequent viable sampling. **B:** In operation, frequent particle monitoring is required. **C, D:** Monitoring on risk basis. Surfaces and personnel should be monitored after critical operations.	**A, B:** Each operating shift for airborne micro, surfaces, and personnel; continuous particulate monitoring. **C, D:** Areas in which products and containers are exposed to the environment: Airborne micro twice per week; airborne particulate once per month; personnel not required. **C, D other:** Airborne micro once per week; airborne particulate once per month; personnel not required.	**A:** Each operating shift. **B:** Each operating shift. **C, D (potential product/container contact):** Twice per week **C, D (no potential product/container contact):** Once per week
Airborne viable action limits (Active air sampling)	**ISO 5:** 1 CFU/m^3 **ISO 7:** 10 CFU/m^3 **ISO 8:** 100 CFU/m^3	Recommends use of Recovery Rate (% of samples with micro contamination) rather than count levels, as follows: Isolator/Closed RABS (ISO 5 or better): <0.1% **ISO 5:** <1% **ISO 7:** <5% **ISO 8:** <10% Applies to all active air, passive air, and surface samples.[5]	**A:** <1 CFU/m^3 **B:** 10 CFU/m^3 **C:** 100 CFU/m^3 **D:** 200 CFU/m^3	**A:** <1 CFU/m^3 **B:** 10 CFU/m^3 **C:** 100 CFU/m^3 **D:** 200 CFU/m^3	**A:** <1 CFU/m^3 **B:** 10 CFU/m^3 **C:** 100 CFU/m^3 **D:** 200 CFU/m^3 0.5 m^3 sample required for A, B 0.2 m^3 sample required for C, D

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(Table 3.0-2 Continued)

Monitoring Guidance	U.S. FDA (Aseptic Processing Guidance)	USP <1116>[3]	EU Annex 1, PIC/S and WHO Annex 4 [2]	Japan MHLW (Aseptic Processing Guidance)	JP XVI
Airborne viable action limits (Passive air sampling)	**ISO 5:** 1 CFU **ISO 7:** 5 CFU **ISO 8:** 50 CFU 90 mm diameter settle plate/4 hr Use of settling plates is optional.	Same sample Recovery Rate as active air, above. 90 mm diameter settle plate/4 hr	**A:** <1 CFU/plate **B:** 5 CFU/ plate **C:** 50 CFU/ plate **D:** 100 CFU/ plate 90 mm diameter settle plate/4 hr	**A:** <1 CFU/ plate **C:** 50 CFU/ plate **B:** 5 CFU/ plate **D:** 100 CFU/ plate 90 mm diameter settle plate/4 hr.	Not specified
Surface viables action limits [CFU][4]	Not specified	Same sample Recovery Rate as active air. Use contact plate or swab.	A: <1 **B:** 5 **C:** 25 **D:** 50 55 mm diameter contact plate	A: <1 **B:** 5 **C:** 25 **D:** 50 24–30 cm^2 contact or swab area	A: <1 **B:** 5 **C:** 25 **D:** 50 24–30 cm^2 (5.4–6.2 cm diameter contact or 25 cm^2 swab area)
Personnel (gown) viables action limits	Not specified. Gown sampling must be established based on job responsibility.	Same sample Recovery Rate as active air.	Not specified	Not specified	Not specified
Personnel (gloves) viables action limits	Not specified	Same sample Recovery Rate as active air.	Glove print, 5 fingers **A:** <1 CFU/glove **B:** 5 CFU/glove	Glove print, 5 fingers **A:** <1 CFU/5 fingers **B:** 5 CFU/5 fingers	Glove print, 5 fingers **A:** <1 CFU/5 fingers **B:** 5 CFU/5 fingers

[1] Guidance is condensed. Refer to the cited references for complete guidance.

[2] Limits are based on averages (currently averaging was removed from the draft of EU Annex 1 (Feb 2020)) – In Grade A no growth is expected

[3] USP utilizes contamination recovery rate instead of excursion limits as found in all other guidance documents.

[4] In general, surface and personnel monitoring should not interfere with the zone protection and should be done after critical operations.

[5] Operators may not be aseptically gowned in ISO 8 support areas.

Note: Settle plates should be exposed for the duration of operations and changed as required after 4 hours (exposure time should be based on qualification including recovery studies and it should not have any negative effect on the suitability of the media used).

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4.0 Environmental Monitoring Qualification of Cleanrooms (Classified Areas) Used for Manufacturing and Filling of Sterile Products

Clean areas for the manufacture of sterile products are classified according to the required characteristics of the environment. Each manufacturing operation requires an appropriate environmental cleanliness level in the operational state to minimize the risks of particulate or microbial contamination of the product or materials being handled.

For new areas, the initial qualification should be performed in a prospective mode. For existing facilities that require a repeat of the initial qualification, the requalification can be performed in concurrence with routine production activities if historical data of existing rooms are shown to be in a good state of control. Release of product should await final release of the cleanroom qualification exercise.

To meet the in-operation conditions, these areas should be designed to reach certain specified air-cleanliness levels in the "at rest" occupancy state. **Section 9.0** (**Appendix 1**) provides additional guidance to ensure and verify that critical system components meet user requirements and are properly identified, installed, and calibrated (typically covered within the installation qualification (IQ) and operational qualification (OQ) phases of cleanroom qualification).

To prepare the cleanroom for classification, air filters should be evaluated for cleanliness and replaced, if necessary, especially in new or modified facilities where excess construction debris or dust may cause clogging. Cleaning and disinfection protocols should be defined and performed prior to conducting qualification. The need to perform baseline viable and total airborne particulate monitoring should be assessed to ensure the cleanroom is ready to go into qualification. This might be a protocol-driven prerequisite.

Performance of the total airborne particulate cleanroom classification described in ISO 14644-1 would confirm the air-cleanliness levels in the at-rest occupancy state. This initial exercise should be supplemented by an at-rest qualification using viable tests.

Viable at-rest locations might focus on more specific aspects related to the actual design of the cleanroom and can be documented as part of the EM risk assessment (see **Section 5.2**). Although human activity and material transfers (the primary sources of microbial contamination) are not evaluated in the at-rest state, the associated sampling locations will provide valuable information about the intrinsic microbial state-of-control of the cleanroom prior to starting the qualification under in-operation conditions. Specifications applied for viable qualification at rest might be more stringent than those applicable for in-operation conditions. All necessary measures should be taken to minimize the impact of personnel during at-rest monitoring. For instance, surface monitoring should be performed at the start of the run, only the EM technician should be present, and delay timers should be applied to automated samplers to allow the technician to leave the location before testing begins.

Subsequently, it should be confirmed that the cleanroom can maintain operational conditions when executing the activities foreseen within the area. The method detailed in ISO 14644-1 can be used to confirm the total airborne particulate cleanroom classification for the in-operation status. This initial qualification is normally performed by simulating manufacturing activities, ensuring that at a certain stage the maximum number of personnel and the worst-case level of activities are covered.

The runs described in this section may form part of the OQ of the cleanroom. Data obtained during this stage should be used as part of the EM risk assessment to define routine sampling locations. For both total airborne particulate runs (in the at-rest and in-operational conditions), one run conducted according to ISO 14644-1 is considered sufficient.

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A performance qualification (PQ) should follow the successful completion of the OQ. However, it may commence before the full OQ results are available, based on preliminary results (business risk approach). If this approach is chosen, PQ protocol should list the results which should be available before the PQ start.

PQ typically consists of three consecutive runs during routine (in-operation) conditions. In batch-related production rooms, a run would typically comprise a batch covering all applicable shifts and activities. In supporting areas, a run should cover all relevant activities to be performed within that area. Simulating activities during performance qualification might reduce the total runtime (i.e., by executing them in a condensed period of time).

The sampling plan should be based on the viable and total airborne particulate sample locations defined as part of the routine monitoring (EM risk assessment) augmented with additional EM locations to cover remote parts of the room in order to confirm and document that the complete cleanroom environment is in control. This PQ can be performed during APS or activities can be simulated ***(10)***. There is a business risk, however, performing PQ during APS since the facility is not operating in a fully qualified environment and deficiencies could be detected that first require corrective measures. Consequently, at a certain stage, the maximum number of personnel and the worst-case level of activities need to be covered during one or more PQ runs. The maximum limits for microbial and total airborne particulate contamination during qualification should correspond to the action limits applied during routine monitoring.

Following the initial PQ, an extended PQ may be conducted for a defined time period based on the routine sampling program or continuation of extended sampling. The intent of this process would be to account for conditions which may not have been covered in the initial PQ (e.g., seasonal variation of microbes, establishment of appropriate alert levels, selection of microorganisms for disinfectant efficacy tests, additional evaluation of the periodic cleaning program, etc.).

Production or room configuration changes with a possible impact on EM (e.g., gowning procedures, cleaning set-up, layout, material or personnel flow, or other areas) should include a new assessment on the qualified status of the cleanroom and may require requalification as part of the change.

5.0 Environmental Monitoring Program

An EM program (including EMPQ) should be in conformance with current GMP, which includes appropriate training and authority of personnel involved in implementing the EM program. Equipment used should be qualified and calibrated, and microbiological growth media should be properly qualified, prepared, and tested. In addition, all operational procedures should be written, approved, and followed with appropriate controls to support their use. The methods selected should be justified and qualified for use as appropriate.

Cleaning and disinfection, of the manufacturing area, and material handling and gowning processes, as well as appropriate aseptic behavior of the personnel, are key components of an effective contamination control strategy.

Selection of sampling methods (see **Section 6.0**), sampling sites, and sampling frequencies should be based on the principles of risk management according to *ICH Q9: Quality Risk Management* to ensure that the respective production area is under control and meets regulatory requirements.

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Establishment of alert levels and action limits may be based on individual sample sites or groups of related sample sites with a similar design or activities. Data obtained are subject to continual review, and alert and action decisions are made by qualified personnel. A documentation system should be in place for identifying and investigating excursions and adverse trends and the effectiveness of the corrective actions taken. All data should be documented and periodically trended.

5.1 Cleaning, Sanitization or Disinfection and Decontamination

EM data demonstrates the effectiveness of the cleaning and disinfection program. The qualification of cleaning and disinfection procedures should demonstrate microbial reduction and maintenance of a microbiological state of control and should provide confidence in the effectiveness of the procedure. This typically includes laboratory surface challenge or carrier studies for contact time and bioburden reduction, supplemented by in-situ studies.

Both USP <1072> and PDA *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities* provide sufficient details of the requirements and in-vitro surface challenge methodology for qualification of the efficacy of the agent used ***(2,15)***.

5.2 EM Sample Site Selection Using Risk Management Principles

Sample site selection, selection of the appropriate sampling methods, sampling volumes, and monitoring frequencies are critical components of contamination control within a facility. However, EM risk assessments are only intended to define the EM program as a part of the overall contamination control strategy. This section addresses how to create and how to maintain an EM program applying a lifecycle approach (**Section 5.2.1**) and the use of quality risk management principles to identify and select sites for EM of classified environments including ISO 5-8/Grades A-D as well as personnel monitoring (**Section 5.2.2**). **Sections 6.8** and **6.9** discuss utilities (WFI/Compressed Air, Nitrogen, etc.).

5.2.1 Lifecycle Approach to an Environmental Monitoring Program

The creation and maintenance of EM programs are best achieved using a lifecycle approach. This lifecycle approach encompasses the evaluation and definition of suitable sample sites, sample types, sampling volumes, and sample frequency for microbiological monitoring that are required before starting up a new facility, area, or room. These sample sites are used to perform an EM performance qualification (PQ). Based on the PQ results, adaptation (mainly reduction or addition) of sample sites may be necessary before setting up the sample plan for routine and, if applicable, batch-related EM. The impact of any major change (e.g., reconstruction, process change, or equipment relocation) needs to be evaluated and the affected area may need to be requalified.

A final EM evaluation may be necessary for decommissioning a room, isolator, or facility to demonstrate environmental control through to the end. The model presented in **Figure 5.2.1-1** outlines the general flow of the lifecycle approach of an EM program. Each step defined is designed to ensure that once an EM program is developed, is it carefully reviewed during the lifecycle of the area, product, or process to affirm that a state of control is maintained. A periodic reevaluation of the EM program (e.g., in response to an adverse trend report) may be necessary.

5.2.2 Initiation of Quality Risk Management Process

Prior to executing a risk assessment, several elements must be defined: scope and objective, risk question or problem statement, risk management tool to be applied, and risk-ranking criteria.

To conduct an EM risk assessment, a risk facilitator should be assigned and a multidisciplinary team should be assembled representing the areas of microbiology, contamination control and hygiene, quality engineering, facilities and utilities, quality control, quality assurance, and manufacturing ***(16)***.

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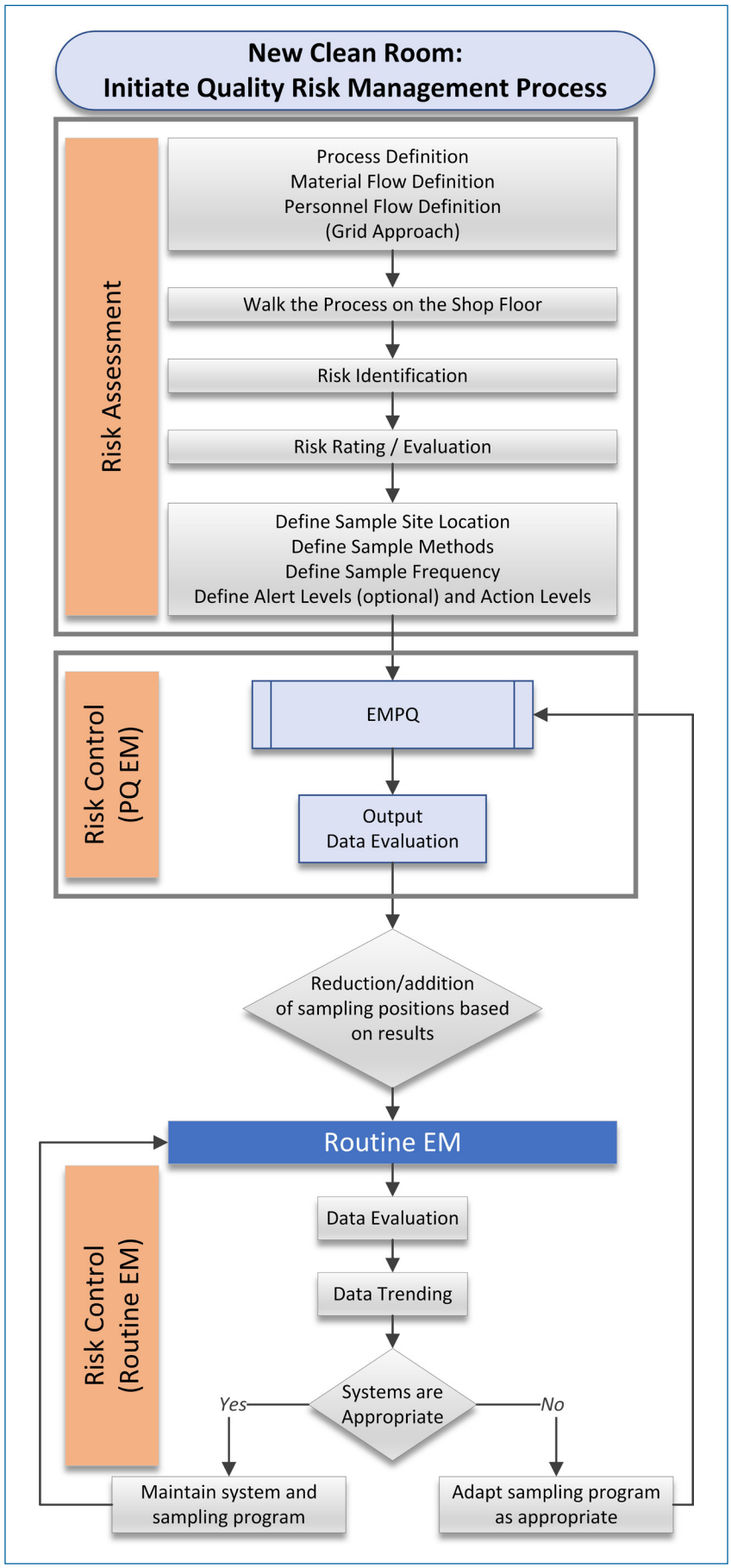

Figure 5.2.1-1 Overview of the Lifecycle Approach to Risk Assessment of an EM Program

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The scope and objective of the risk assessment is defined as the area of the facility or process under assessment of the EM program. The scope can involve just one room, an isolator, or a larger manufacturing area. Independent of the scope, however, the manufacturing process associated with the area in scope is key to performing the risk assessment.

Thorough process understanding and process mapping is necessary to evaluate the associated risk of contamination at a specific position within a cleanroom. Prior to initiating the risk assessment, process mapping should be conducted on the shop/manufacturing floor using room layouts and indicating equipment locations, people flow, material flow, and waste flow in the facility. During this process mock-up, the team will identify potential hazards within the areas under assessment.

The assessment team will select the most appropriate risk management tool to be used among the different methods applicable, e.g., Failure Mode and Effects Analysis (FMEA), Hazard Analysis Critical Control Point (HACCP), among others ***(17)***. One risk assessment tool, developed by a collaborative of pharmaceutical industry representatives through BioPhorum, addresses the wide range of practices within the industry, allowing harmonization as well as compliance with regulatory guidelines. This toolkit provides an easy to use, step-by-step guidance for establishing EM programs ***(18)***.

For process mapping, grid profiling can be useful to describe single process steps and consider activities specific to that grid. After gridding the room layout, the process steps in each grid are evaluated and the risk ranking is determined based on six risk factors: 1) cleaning and sanitization access, 2) personnel presence and flow, 3) material flow, 4) proximity to open product or product contact material, 5) interventions/operations by personnel and their complexity, 6) frequency of interventions (see **Appendix 2 Figure 10.0.1**).

Based on the risk scoring, sampling locations are determined for performing the EMPQ and, afterwards, for conducting the routine monitoring. To enable systematic assessment of all cleanroom areas, a recommendation of the grid size for the respective grades is made (e.g., 0.5 m^2 for Grade A). In addition, grouping grids into functional sections (e.g., filling area) enables meaningful sample points to be defined. Various risk assessment examples from industry are available illustrating this approach ***(19)***.

Depending on the tool selected to execute the assessment, the risk-ranking elements must first be defined. To perform a multidimensional risk management approach, a minimum of two risk-ranking elements (e.g., proximity, complexity, frequency, severity) must be defined. Examples of risk assessments using various tools are provided in **Sections 9.0-11.0** (**Appendices 1-3**).

5.2.2.1 General Points to Consider for a Risk-Based EM Program (Risk/Hazard Identification)

Sample locations should be chosen based on the criticality of the process being performed in the room or area, product exposure to the environment and susceptibility to microbial and particulate contamination (i.e., serve as the most sensitive measurement of the state of control of the room), locations of potential ingress from supporting areas or adjacent non-classified areas (i.e., gowning areas, airlocks, corridors), and sites shown to be appropriately placed to detect a problem in the area.

Suitable sample sites will depend on the area design and manufacturing process. Locations posing the most critical microbiological risk to the product are a key part of the program.

The following factors should be considered when the team is performing the walk-through of the facility:

1. Adherence to industry and regulatory guidelines, e.g., CFR, ISO 14644, USP, FDA, EU (**Note:** ISO 14644 only provides a guidance for the classification of cleanrooms, e.g., minimum number of sample locations for total particulates)

2. Sites and locations where microbial and particulate contamination would most likely have an adverse effect on product quality and, therefore, have the highest risk, considering—

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- Proximity to open product or product contact surfaces and critical sites (e.g., filling needles, stopper bowls)
- Activities linked with interventions
- Areas that are the most inaccessible or difficult areas to clean and disinfect
- Locations with a high frequency and/or complexity of activities by cleanroom operators, (e.g., touch panel, forceps, door handles)
- Areas with a large number of personnel and high personnel flow (e.g., floors at the entrance of gowning rooms)
- Areas with high material flow

3. Uniform geometric pattern or grid-profiling within the cleanrooms, to cover the complete area
4. Assessment and a justification for locations that will not be part of the EM program due to certain restrictions or alternative/worst-case coverage
5. Historical data and/or data obtained during qualification
6. Anaerobic organisms:
   - Strict anaerobic strains are unlikely to be recovered in common aerobic cleanroom environments. Periodic and/or investigational monitoring for anaerobic organisms may be conducted as a risk-based activity; it is mainly performed to detect any micro-aerophilic bacteria, such as *Cutibacterium* (formerly *Propionibacterium*) *acnes* that usually are not detectable applying standard aerobic monitoring conditions.
   - Generally, a flush or overlay with nitrogen or carbon dioxide does not deplete or replace enough oxygen to create anaerobic conditions that require routine anaerobic-viable monitoring of the area; however, periodic testing of the compressed gasses utilized must be considered.***(11,14,15)***
   - The frequency of anaerobic testing may differ from the routine monitoring schedule. Although anaerobes are a significant component of the skin microbiota, aerobic environmental monitoring is considered adequate to detect inadequate gowning and poor aseptic technique by cleanroom personnel. Anaerobic environmental monitoring in response to a sterility test failure implicating strict anaerobes should be considered.

7. Airflow visualization studies, that is, smoke studies; areas of concern, such as turbulences and eddies, may be addressed through airflow, process, or facility redesign.

5.2.2.2 Risk Initiation: Development of EM Risk Assessments

Typical approaches to EM risk assessments include the following activities:

- Map the layout of the room(s), overlay with grids, and combine them into functional sections
- Walk the process with the risk assessment team, detailing process activities grid-by-grid
- Assess each grid against the risk factors and score them to determine the relative probability of contamination
- Evaluate the results by functional sections to select sampling locations and methods
- Create an EMPQ plan
- Evaluate the EMPQ results
- Determine routine EM sampling locations, methods, and frequencies
- Review the risk assessment on a defined, periodic basis

For each grid, risk is scored based on the process used, and then ranked based on these six factors:

1. Accessibility of equipment and surfaces for cleaning and sanitization
2. Personnel presence and flow
3. Material flow
4. Proximity to open product or open product-contact materials
5. Need for interventions/operations by personnel and their complexity
6. Frequency of interventions and operations

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Strategies regarding risk scoring and risk ranking can be found in the BioPhorum EM Risk Assessment Toolkit.***(18)*** For example, each risk factor is ranked as low (1), medium (2) or high risk (4), with exception of "proximity to product," which is ranked as low (1), medium (4) or high risk (8). The calculated product for each grid results in the risk scoring as low risk (score of 1–16), medium risk (score of 32–265) or high risk (score of 512–8192). A detailed example can be found in **Section 10.0** (Appendix 2).

5.2.2.3 Risk Assessment

Once the elements from of the EM Risk assessment are defined, the team will begin the risk assessment phase of the lifecycle. Next, the hazards identified during the facility walk-through will be documented, and the hazards will be evaluated for risk factors given the current controls in place. During this phase of the risk assessment, all current controls are accounted for and documented, and consider the following items:

- Timing of sampling and sample volume measured against the company standard
- Frequency of sampling measured against the company standard
- Additional criteria (e.g., height of sampling location)
- Risk of sampling procedure for product contamination (part of the justification if a site is not sampled)
- Locations and areas to be assessed where there is no monitoring
- Historical data
- Air flow studies

Once the risk ranking is completed, the sampling site selection and sampling frequency can be determined. Additionally, if risks are identified which are not within the organization's risk tolerance level, those risks must be mitigated.

After having considered the relevant criteria and factors, in the case of a new EM program, a rough plan and site layout (with locations) should be prepared; or, in case of existing sample locations, these locations can be assessed against the relevant criteria.

The outcome of the risk assessment includes a documented plan for monitoring and the rationale for the EM program (i.e., sampling locations, sampling volume, sampling frequency, methods). With a new EM program, there will be little historical data available to inform the risk assessment. In these cases, the team will need to focus on the factors that may contribute to or perpetuate contamination, using scientific experience and subject matter expertise to minimize the uncertainty in the assessment. Over time, the data collected during routine monitoring will reveal the risk level.

If there is an existing EM program without a risk-based rationale to support the EM program, the risk team must be careful to ensure that they are not performing an assessment to justify the current sampling program. In these cases, the team must evaluate the facility/process (e.g., via a Gemba walk), taking into consideration historical data to prevent removal of sampling locations which have historically demonstrated adverse recovery trends. **Note:** A Gemba (Japanese) walk is a practice of Lean Six Sigma, to watch how a process is actually performed and talking with those who actually perform the work.

Where an existing EM program was created using the principles of risk management, the team will evaluate the existing assessment to ensure that all conditions assumed in the original assessment are still in place. If there are discrepancies between the initial and current conditions, the assessment is modified, and the hazards are re-ranked to determine if the risks are below the required threshold. Any major deviations to the relevant criteria should be considered, and mitigation actions may be defined to control the risks.

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5.2.2.4 Risk Control and Risk Review

Throughout the lifecycle of the EM program, if risks are found to be above the allowable threshold, the team will develop risk control actions, which will lessen the likelihood of contamination occurring and/ or increase the likelihood of detection by defining additional sample locations. Once these additional controls have been implemented, the assessment will be revisited to ensure the controls were effective in minimizing the risk. In addition to reviewing the effectiveness of in-process or procedural controls, the completed risk assessment is subject to review on a routine basis (e.g., annually) to ensure the EM program is functioning as designed.

5.3 Alert Levels and Action Limits

EM programs require action limits to be established based on the applicable regulatory guidelines or requirements. These guidelines frequently recommend alert levels also be established. Some companies also choose to set levels for individual cleanrooms or sample sites. The alert level will be established based on statistical analysis of the historic EM data.

Alert levels and action limits based on CFU count excursions have been eliminated from USP <1116>, which indicates a paradigm shift to contamination recovery rate (CRR).***(9)*** CRR is the percentage of environmental samples found to have microbial contamination (e.g., a CRR of 1% would mean that only 1% of the samples taken showed contamination, regardless of colony numbers). The CRR approach may lend itself to wider applicability considering emerging EM technologies, such as those not reliant on CFU measurement. However, current official regulatory guidances from both the EU and the US still retain GMP requirements for alert levels and action limits. Consequently, companies may want to voluntarily monitor CRR as an EM trending tool in addition to the required alert levels and action limits.

The response to alert levels or action limits should be written and employed consistently. To create consistency, investigatory and corrective action steps should be specified in advance. Records should show that the excursion was recognized, appropriate follow-up occurred, and appropriate corrective and preventive actions (CAPA) were taken. Established alert levels, should be reviewed periodically as part of routine trend analysis. They may be revised to reflect improvements, advances in technology, changes in use patterns, or other changes.

When no regulatory or industry guidelines are provided, alert levels and action limits may be derived statistically from historical data. Statistically derived alert levels, based on performance of the area, might be low; however, one should be using those values to detect atypical recurrences. Other considerations in adjusting levels include improving process capability, consistency of alert levels and action limits for similar room classifications, level of gowning, and product contamination risk. Infrequent excursions from these levels are expected at frequencies characteristic for the specific mathematical model used in their derivation. The critical requirement is recognizing an adverse trend that may represent a loss of environmental control.

The alert levels and action limits do not define product attributes, such as sterility, and should not be considered as product specifications. Rather, alert levels are intended to indicate changes or drifts in environmental control so that corrective action may be taken before product quality is adversely affected. However, where an action limit is exceeded where critical aseptic process steps are conducted, the investigation should be completed prior to release of the batches impacted by the excursion, which should include an evaluation of the product impact, based on the most probable root cause identified.

Since there is no one best method for setting alert levels, the approaches described in **Sections 5.3.1–5.3.3** may be taken within the pharmaceutical industry. Where compendial requirements exist, as with action limits, they supersede these examples.

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Using standard statistical tools for setting alert levels and action limits can be misleading because environmental monitoring data is not normally distributed. No pharmaceutical industry or regulatory consensus exists on the best statistical model to use for setting these levels. Therefore, it may be appropriate to determine the model that best fits the data and use that model to set the levels. When data does not fall within a normal distribution and exhibits high levels of skewness towards lower counts and /or zero counts, a non-parametric approach to setting alert levels and action limits are suitable when large number of historical observations exists. In the absence of large data sets, non-parametric approaches run the risk of selecting as the control limits, the outlying data points caused by out-of-control conditions, resulting in inflated estimates of alert levels and action limits.

Traditionally, alert levels and action limits are established, using a mean of 2 or 3 standard deviations (SDs) to render certain degrees of assurance that the classified environments are in control. This method hinges on the assumption that the data or data after a transformation such as logarithmic are normally distributed.

Some of the commonly practiced approaches are described below.

5.3.1 Nonparametric Approach

A nonparametric evaluation is based on 99.9th and 99.99th percentiles and can be chosen as the alert and action levels. These levels correspond to about mean plus 3 and 4 sigma ranges, respectively. These levels are commonly selected as process alert and action levels. When the requirement of proper sample size is met (N≈1000), the alert levels analysis establishes confidence that during actual process monitoring 99.9% of the population will lie below the stated value. When these limits are exceeded, it will trigger investigation that will prevent contamination or an adverse trend condition.

Action levels can be selected at 99.99th percentile. The action level analysis establishes confidence that during actual process monitoring 99.99% of the population will lie below the stated value. Therefore, the derived action levels serve to identify an excursion at the selected percentile level. By definition, it is very unlikely that validated system, operating in a state of control, will deliver results that will deviate significantly from the norm especially when multiple alert/action excursions in same room in the same day are detected. This condition is likely indicative of a significant drift the process control(s) or a systemic failure of the validated system has occurred.

5.3.2 Nonparametric Tolerance Limit Approach

As environmental monitoring data, especially in cleanrooms, is typically not normally distributed (i.e., exhibits skew heavily toward lower counts or zero counts), a nonparametric tolerance limit approach to setting alert levels and action limits is recommended. These limits allow the user to assert with at least 95% (K=0.95) confidence that 100(P) or 99% of a population lies below the value, as determined by the stated action limits, for the respective data.***(16)*** For distribution-free tolerance limits, minimum sample sizes are N=60 for 95/95 (alert level) and N=300 for 95/99 (action limits).

5.3.3 Cutoff Value Approach

All the test data for a particular site, or group of similar sites, are arranged in a histogram, and the alert levels and action limits are set at values whose monitoring results are respectively 5% and 1% higher than the level selected (i.e., 95th and 99th percentile, respectively). Other percentiles may be used in establishing levels. Alert levels are typically determined per test type (active air, passive air, and surface monitoring).

The alert definition should be based on a timeframe allowing the accumulation of enough data to be representative of a population. A general text on statistics (*Juran's Quality Handbook, The complete Guide to Performance Excellence, Joseph A. Defeo, 7th Ed. McGraw Hill Education Books, 2017*) recommends a statistically significant value of at least 20 data points. Alert levels are typically set at about 50-60% of action until

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enough data is generated. An example is to take the last 100 monitoring results or data collected during one full year and use the 95th and 99th percentile values as the alert levels and action limits

Areas that have the same activities may be grouped to provide more data points; for example, material airlocks of the same Grade, or Grade D hallways of same function can be grouped.

The calculated alert level should be challenged for its appropriateness, however, avoid modifying or increasing the alert as such it loses any correlation with your actual cleanroom performance. In addition, performance-based alert levels that seem to be well below the applied action limits should be considered as a confirmation of a solid microbial control of the environment and does not per definition to be considered as inappropriate. One should be aware that above alerts will occur and that this is inherent with the use of a cut-off approach. One should be more focused on the level of occurrence (see its use as a trigger for trend analysis within **Section 5.4.3**).

If the outcome of the periodic reassessment of the alert assessment results in a level increase, this should be interpreted as a warning that the facility may be slowly drifting out of control or might be a result of modifications of the activities within the area (e.g., increase in batches produced, changes in process). It is recommended not to increase alert levels without a good understanding of the reasons behind it.

The frequency-based excursion rates and trends can be used to determine the state of the EM control and gowning process. The cumulative conforming count chart (CCC) is easy to use and more sensitive to short-term drifts in process parameter, enabling its user to judge the state of control. The idea behind the CCC is the fact that the number of conforming items between two consecutive nonconforming items changes when the fraction nonconforming shifts is observed. CCC plots the number of inspectional monitoring data sets between excursions. The method is suitable when large number of data exists.

At a specified excursion rate, a likely number of excursions per given number of samples can be estimated at a 95% confidence level. The CCC can be used to identify the average and the minimum number of continuous successful (within alert levels) facility- or gown-monitoring examinations before the data shifts. Such approaches can be used to evaluate limits, define trends, and assess training effectiveness, for example, gowning process and operator or analyst disqualification and requalification.

Other models based on negative binomial, Poisson, Weibull, or exponential distributions are possible. It may be appropriate to determine the model that best fits the data and use that model to set the levels. As noted, contamination in strictly controlled environments does not typically fall within a normal distribution. Environmental monitoring data may be evaluated to determine the suitability of the approaches to level setting.

The monitoring group should review the data for trends at an appropriate frequency. The quality unit should review quarterly and yearly monitoring reports.

5.3.4 Quantitative and Qualitative Trending Rules

Trend rules should supplement alert levels and action limits and focus on two important axes:

- To rapidly detect gradual bioburden changes before it leads to real excursions with potential impact for production and/or the cleanroom environment (immediate term trending)
- It should be able to periodically evaluate the general state of control of the cleanroom environment (short- and long-term trending)

A useful trending rule is to look at alert level excursions over time within the same room identifying correlation between/among locations, or frequent sustained recoveries above alert. When trend rules are exceeded, an investigation is recommended.

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Immediate — short term trending rules typically look at an increase in the proportion of above alerts over a specified time period. Examples:

- A specified number of above alerts at a single location/type of test
- A specified number of alerts within one room (same and different types of tests)

Some statistical rules can be used as well in order to identify when the proportion of above alerts is moving to an atypical level.

Short Term — monthly or quarterly

The monitoring group should perform a comprehensive review of the data at an appropriate frequency as a function of the amount and criticality of the data generated. The quality unit should review and approve the periodic monitoring reports.

Typically, a thorough analysis with a statement if data indicate that the facility is operating in a state of control or if data indicate some issues, how it was discovered, where and what corrective action was taken and what impact the issue had on the facility. The integration of graphs is highly recommended to visualize the state of control.

One can assess:

- Same sample type from same location exceeding the action limit multiple times
- Increases of the number of alert level excursions (e.g., same monitoring type: active, passive or surface monitoring)
- Recovery rate increases for the same location (e.g., all personnel on one filling line), or the same room or area.

Long Term — annual

Annual trending should report the changes in a year's worth of data. Since it incorporates more data and thus is should be able to give a better holistic picture of the cleanroom environment.

It might include:

- Number of EM Deviations per grade with indication of concerned rooms and test type and corrective actions
- Occurrence of atypical microorganism per grade and explanation of potential root cause identified
- Microbial flora analysis and confirmation of the disinfection program efficiency and validated state
- Global analysis of the site performance covered by the monthly/ quarterly reports and comparison with the previous period
- Key findings and highlights on any change in SOP or modification during the year
- Year graphical representation and global analysis of the data from your cleanrooms and conclusion with a statement such as data indicate that the facility is operating in a state of control or if data indicate some issues and any related CAPAs
- Analysis of potential correlations between observed product contamination (e.g., bioburden) and the micro-organisms recovered within the cleanroom environment.

5.4 Data Management

Routine review and analysis of EM data for trends at an appropriate frequency is essential to aid in the interpretation of process stability and assess overall environmental control performance. Management must be kept abreast of trends and the subsequent state of operations within facilities with review of quarterly and yearly monitoring reports.

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Based on the large number of samples tested by a given facility, a computer-based data-tracking system is essential. Before implementation, all database applications used should be validated or qualified for specific software applications.

5.4.1 Data Integrity

The PDA *Elements of a Code of Conduct for Data Integrity in the Pharmaceutical Industry* ***(19)*** and PDA *Technical Report No. 80: Data Integrity Management Systems for Pharmaceutical Laboratories* ***(20)*** provide comprehensive information on this topic.

Collecting, analyzing, reporting and retention of EM data should be done under conditions that are designed to guarantee reliability and integrity of the data. The data are required to be attributable, legible, contemporaneous, original, and accurate (ALCOA):

Attributable: Information is captured that identifies where the data came from

Legible: Data are readable and understandable by humans

Contemporaneous: Data are recorded at the time they are generated or observed

Original: Data are recorded for the first time

Accurate: Data are correct

Important elements to consider with respect to environmental monitoring are:

- Integrity of the sampling site data by including a formal mechanism to enable full reconciliation of sample collection, incubation, and evaluation
- EM data must be documented on controlled worksheets for paper records or entered into a validated computer system with appropriate security, audit trails, validation, and oversight
- Employees must adhere to the established site documentation requirement and will not record raw data on unofficial forms, writing pads, or other uncontrolled media
- Employees will not discard or destroy raw data or original records, other than at the end of the prescribed retention period

The deployment of a second-person verification should be considered to confirm the enumeration of microbial recovery. The main focus, however, should be to have in place a robust plate-reading process, reducing the risk for variability to the minimum, for instance, by applying appropriate lighting and/or magnification and a robust training of the technicians performing the readout. The utilization of QRM principles can aid in the determination of whether or not a secondary verification is required (e.g., Grade A EM). Automated systems for EM test readouts and data storage can eliminate bias and the need to perform a second verification.

Routine data are aligned into a source in a consistent record format. At a minimum, the record format should include the following:

- Identification of the operator and analyst
- Monitoring date and time
- Equipment used
- Specific sampling locations
- Sampling methods including media used
- Incubation conditions
- Colony-forming units (CFU) or total airborne particulate count results
- Identifications performed
- Batch information (if applicable, e.g., for batch-related monitoring)
- Current alert levels or action limits

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Records must be signed and verified by the appropriate person, depending on the type of system used. Some alternative microbiological methods can use measurements other than CFUs, provided they have been properly validated before use, for example, relative light units, cells, etc. A manual data entry or image scanner system, with the advantages of speed and accuracy, can be used to populate tables. Regardless of the type of system used, data integrity must be verified prior to data collection and analysis.

5.4.2 Data Trending Analysis

Histograms or tables, characterized by a number of data points that fall within a common frequency, are valuable tools to conduct the trending expected by regulatory agencies. Different room classifications with defined requirements will produce different histograms. For example, the CFU spread obtained across an ISO 8 data set will not be observed in a data set from an ISO 5 area. Therefore, each area, or area type, and its accompanying data set must be viewed as distinct. Data trends in counts, occurrences, and organism types can provide a holistic view of the cleanroom environment. They can also be used to evaluate if there is ingress from lower to higher grades. Cleanroom maps indicating sampling excursions are useful as well to show connectivity of various grades and potential ingress through equipment and personnel flows. While this does not require statistical tools, mathematical models could be applied. Examples of statistical methods and control charts can be found in PDA *Technical Report 59: Utilization of Statistical Methods for Production Monitoring* ***(21)***.

5.4.3 Interpretation

Routine environmental microbial monitoring data should demonstrate that the classified area is operating under an adequate microbial control for the needs of the operations conducted in that area. Individual recoveries in critical areas (i.e., Grade A) must be adequately investigated. Compliant EM results do not guarantee the sterility assurance of the process, however, as EM incorporates multiple limitations (e.g., recovery efficiency of the applied methods, selected sample locations etc.). EM can only confirm to some extent the robustness of the design and processes.

Data generated should be summarized and evaluated to determine whether the EM process is in a state of control and documented in periodic reports. A variety of methods are available to perform this analysis, for example, the use of frequency distribution and control charts or statistical analysis.

Measurements for viable contamination, such as contamination excursion rates and/or recovery rates, can be used to determine the level of control in a given area. The excursion rate is related to the number of samples exceeding the defined alert levels and action limits, whereas the contamination recovery rate (expressed as a percentage) is related to the number of samples that are positive for microbial growth when compared to the total number of samples taken within the same classified area(s). Therefore, recovery rates differ from excursion rates. Given that microorganisms are not homogeneously distributed in the same environments, and the sensitivity and variability associated with microbial sampling methods, these rates can be useful approaches to trending results in controlled areas. The combined excursion rates are calculated by determining the number of samples with excursions outside of established levels, dividing by the total number of samples collected, and converting to a percentage. **Table 3.0-2** shows the recommended acceptable action limits and recovery rates for various ISO classifications. For an ISO 5 environment, an excursion rate of not more than 1% is achievable due to stringent controls, and the expectation is that the results show no viable recoveries.

In addition to individual excursions, CRRs may be used for trending and control of the overall microbial load in the classified environment, equipment surfaces, material, and personnel. **Table 5.4.4-1** summarizes CRRs recommended by USP <1116> for various classified areas. Companies can utilize the criteria from USP <1116> to develop their recovery rates for cleanroom environments depending on the activities and processes conducted in those areas. They should also establish a CRR for each monitoring attribute, such as surfaces, air, and garments, for each classified area. Any changes from

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the established calculated contamination rates should be investigated and corrected. However, any contamination in Grade A/ISO 5 areas must be investigated, since there should not be any microbial recovery within those areas. The CRR provides an alternative measurement of the effectiveness of the overall contamination control strategy.

Table 5.4.4-1 Contamination Recovery Rates Suggested by USP Chapter <1116>

Room Classification	Suggested Initial Contamination Recovery Rates (%)			
	Active Air Sample	Settle Plate (9 cm) 4-Hour Exposure	Contact Plate or Swab	Glove or Garment
Isolator/closed RABS or ISO 5 or better	<0.1	<0.1	<0.1	<0.1
ISO 5	<1	<1	<1	<1
ISO 6	<3	<3	<3	<3
ISO 7	<5	<5	<5	<5
ISO 8	<10	<10	<10	<10

Note: All operators are aseptically gowned in these environments. These recommendations do not apply to production areas for nonsterile products or other classified environments in which fully aseptic gowns are not donned. In those conditions other Recovery Rates might be used if appropriately justified.

Evaluating the recovery rate levels at a defined frequency is recommended (at least monthly as per USP <1116>). Trends may include a gradual increase or decrease in the overall counts observed over time, or a change in flora or counts on several plates of a particular area on a given day. Interpretation of the impact of a significant fluctuation in counts or a change in flora should be based on the experienced judgment of a qualified microbiologist.

Some considerations for assessing the state of control for a given process include:

- Multiple excursions observed within a specified time period may be considered a pattern or cluster formation that, if above the alert level, signals a trend that requires investigation. As an Alternative, such approaches as statistically derived frequency of excursion can be used and justified.
- Fluctuations in environmental monitoring data values for a given process or a specific area require further attention as this may point to construction activities, facility conditions, change in process, flow of material, seasonal variations, disruptive events, or other factors.

Understanding the potential impact of the results generated during EM is critical to a successful EM program.

5.5 Rapid Microbiological Methods in Environmental Monitoring

The majority of EM performed currently relies on conventional methods based on the recovery and growth of microorganisms using solid or liquid microbiological culture media. However, these methods are often limited by slow microbial growth rates, long incubation periods, unintended selectivity of culture media, and the inherent variability of microorganism recovery rates due to their non-specific response to culture methods. Rapid methods that provide faster time-to-result may offer an advantage for EM tests over current, conventional approaches.

Rapid methods may afford the industry with several opportunities when applied to EM:

- Confirmation (in real-time or close to real-time) of microbial control during manufacturing activities, including manual aseptic manipulations, filling, and packaging

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- Confirmation of microbial control in controlled areas prior to use (e.g., manufacturing suites, gowning rooms)
- Rapid assessment of controlled areas during facility qualification activities (e.g., EM PQ)
- Evaluation of contamination remediation efforts and root cause analysis (e.g., using rapid methods to find a source of contamination and/or determine the effectiveness of cleaning and disinfection procedures)

Subsequently, the industry should look toward utilizing innovative approaches for the detection and quantification of microorganisms as part of a holistic EM program. From a QRM perspective, the benefits of implementing rapid microbiological testing strategies can help to design robust processes that prevent contamination, ensure that a state of microbial control is maintained, evaluate the impact of human interventions during aseptic filling, establish recovery time after power shortages, develop more effective strategies to correct a contamination problem, continually improve processes and products, and quickly and efficiently assess the potential impact of failing results on the patient.

Additional, detailed guidance related to **Sections 5.7.1–5.7.5** may be found in PDA *Technical Report No. 33: Evaluation, Validation and Implementation of Alternative and Rapid Microbiological Methods* ***(22)***.

5.5.1 Rapid Microbiological Method Scientific Principles Applied to Environmental Monitoring

Rapid methods appropriate for EM are based on a wide variety of scientific principles. Some methods still require growth of microorganisms using conventional microbiological media; however, detection and enumeration can be faster if using novel detection techniques, such as:

- Digital imaging and autofluorescence of microcolonies
- Fluorescent staining and laser excitation of microcolonies
- Rapid detection of carbon dioxide (CO_2)
- Detection of adenosine triphosphate (ATP) after a brief microbial enrichment period

Other rapid methods do not require microbial growth prior to detection or enumeration and, as such, the time to result can be instantaneous or close to real-time. Rapid technologies that achieve these results include:

- Flow and solid-phase cytometry using viability stains and laser detection systems
- ATP bioluminescence, when enrichment is not required due to the concentration of microorganisms in the sample
- Intrinsic fluorescence and optical spectroscopy detection of microorganisms in air or water
- RAMAN spectroscopy coupled with viability staining
- Nucleic acid amplification techniques, when enrichment is not required due to the concentration of microorganisms in the sample

The most appropriate technology and detection method to consider will depend on a variety of factors, including, but not limited to:

- Sample type
- Sample size required for analysis
- Sample compatibility with the rapid method workflow and instrumentation
- Desired time to result
- Level of sensitivity required
- Need to neutralize antimicrobial activity for methods requiring growth
- Ability of the rapid method to allow identification of any recoveries (destructive vs. non-destructive assay)
- Potential for false positives or false negatives due to cleaning or disinfectant residues

Automated and high-throughput methods using conventional media are also available that may afford faster results with better efficiency in terms of automation, and data management and data integrity.

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5.5.2 Benefits of Using Rapid Microbiological Methods

The implementation of rapid methods for EM may present the industry operational enhancements related to product quality, process knowledge, and regulatory acceptance, such as:

- Potential for reduced in-process microbiology testing and finished product release cycle times
- Reduction in risks associated with forward processing
- Elimination or reduction of off-line microbiology assays
- Increases in laboratory automation and reductions in manual testing, sample handling, and/or data management and data integrity
- Reduced overhead and/or headcount for sampling and/or testing
- Ability to make immediate quality decisions on the state of microbial control
- Faster response to contamination events and initiation of investigations
- Reduced repeat testing, lot rejection, reprocessing, and rework
- Reduction in room or facility downtime and investigations
- Faster confirmation of contamination remediation efforts

5.5.3 Validation of Rapid Microbiology Methods and Equipment

Rapid microbiology methods vary in complexity and in the associated equipment required. The validation of software and equipment follows the equipment validation model as found in USP <1058 >; the validation of the methods is discussed in compendial chapters of U.S. and EU pharmacopeias, USP <1116> and Ph. Eur. 5.1.6. respectively. ***(9, 23, 24)*** PDA *Technical Report No. 33* also provides industry-proven strategies that may be employed during validation. ***(22)***

The most recent public draft of EU Annex 1 (Feb 2020) recommends that when a different or new microbiology technology is used that presents a result in a manner different from the CFU, the manufacturer should scientifically justify the limits applied and, where possible, correlate them to the CFU (see **Table 3.0-2**, footnote #2). This is true for Grade A areas in which "no growth" is expected from active air, passive air, and surface samples during the qualification of the area and routine monitoring.

The use of a rapid method for EM may result in data that can be compared with existing alert levels and action limits, especially when the rapid method relies on the growth of microorganisms and the data signal is the CFU. If the data signal is not a CFU, however, the possibility exists that the resulting quantitative result will not align with existing alert levels and action limits, even though a correlation between the CFU and the new signal can be made. In this instance, the end-user may elect to generate sufficient data to modify the existing alert levels and action limits based on historical data associated with the new method and data signal. The new levels that are applied to the rapid method should be rationalized and meaningful, and a risk-based analysis or approach should be used to finalize the setpoints.

Alternatively, trending the EM values may be more appropriate than looking at individual counts, as this approach may be more suitable for determining when a microbiological problem is occurring. This may be especially apt when performing EM in well-controlled environments in which zero cell counts are expected (e.g., Grade A (ISO 5) areas).

Results obtained using a rapid method may differ from traditional methods that have been used for routine EM. Traditional EM utilizes standard nutrient agar plates and depends on a number of factors (i.e., time, temperature, humidity, nutrients) in order to recover microorganisms. Many rapid methods do not require recovery and growth of the microorganisms to generate a quantitative result. Therefore, it may not be possible to draw a direct correlation between the number of CFU generated from an EM sample compared to the result generated by a rapid method during side-by-side equivalence testing. In this case, additional statistical analyses between the two data sets may be warranted. Furthermore, recovery may be numerically higher using a rapid method that does not rely on microbial growth, as organisms that would not be detected using a traditional, growth-based method could be detected using alternative methods. This is especially true when the organisms are viable, but not culturable (VBNC), physically stressed, damaged, or in a dormant state.

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5.6 Identification of Isolates

Identifying and strain-typing of microorganisms is not only important as part of the cleanroom qualification or when related to EM excursions but should also form an integral part of routine surveillance monitoring programs.

The identification program selected by the laboratory should be defined in writing, including the frequency of identification, standard procedures for the methods used, and consistency with regulatory expectations for identifications. Companies that rely on the services of contract laboratories for microbial identification should qualify them per the requirements of their own supplier qualification procedures and have a quality agreement in place.

The identification of microorganisms recovered within the different cleanroom environments forms an important element to assess its microbial state of control, allowing:

- Detection of any microbial changes and potential seasonal variations of the flora
- Definition of potential sources of the microbial isolates related to a detected contamination (e.g., as part of an investigation)
- Assessment of the level of contamination control
- Response upon recovery of undesirable organisms that might pose particular risks

The Grade A environment is expected to be completely free of recovery of viable organisms because any presence of an organism within this area is considered undesirable. Within its surrounding environment (e.g., Grade B), an additional challenge might come from organisms that are spore forming (mold species, *Bacillus* species) and put additional stress on the Grade A/B barrier. In this context, water-borne Gram-negative microorganisms should be considered undesirable and of particular concern since the use of nonsterile water sources is not authorized within aseptic cleanrooms. Therefore, presence of Gram-negative microorganisms may indicate water intrusion which would also increase the potential risk for mold proliferation.

In Grade C and D environments, levels of undesirable organisms should be controlled in order to limit any microbial stress on surrounding aseptic cleanrooms.

Table 5.6-1 illustrates an example of a scheme for the extent of identification that may be used for the recovered microbial isolates. The extent of identification and rationale should be documented and should be determined on a case-by-case basis, risk assessment, facility validation, and appropriate trend analysis.

Table 5.6-1 Typical Scheme for Microbial Identification

Extent of Identification	Isolate and Origin
Identification to species (or genus where speciation is not possible)	All recoveries in Grades A & B Grades C & D, for alert level or action limit excursions Periodically as part of the ongoing surveillance program
Strain typing or molecular fingerprinting	Significant product failure (e.g.) • media fills • sterility test • significant adverse trends

For some types of processes and products, concern regarding specific organisms may determine the level of identification required.

Initially, many isolates may be identified to establish a database of the representative microorganisms found in the area. Periodic identifications should be performed on routine monitoring to check for any changes in the predominant microflora, even if no excursions have occurred in the Grades C and D ar-

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eas. A change in the microflora might signify a change in the environmental controls in the facility that should be investigated. Identification can be useful information as to the possible source of isolates. For example, *Staphylococcus* species are commonly found on skin, *Pseudomonas* species are usually associated with water sources, and *Bacillus* species and mold are generally connected to soil, dust, and air.

5.7 Investigations and Corrective Actions

Investigations are needed in response to an action-limit excursion and to address an adverse trend in order to determine a cause-and-effect relationship or most probable cause (i.e., sources of contamination). For consistency in the response to excursions, investigative and corrective action steps should be specified in a written procedure (see **Table 5.9-1**). The procedure should define the level of investigation required if there are multiple or sequential excursions. The investigation should include a product impact assessment and evaluate the risk to other batches of the same or related products manufactured in the same time frame. Critical investigation should be closed prior to product release.

Table 5.7-1 Examples of Investigation Elements for Different Systems

System	Investigation Elements
Compressed gas system	Repeat test immediately. Perform filter integrity testing.
	Replace filter if excursion is confirmed on retest.
	Evaluate impact on processed component and/or product.
Room air/ HVAC	Review level of personnel activity.
	Review/perform airflow patterns and HEPA integrity tests.
	Review aseptic technique of personnel and training records.
	Review gowning procedures and requirements for area.
	Review trends and any incidents of HVAC outages, if any occurred.
	Inspect incoming air filters for leaks and pressure differential across filter.
	Review room cleaning and disinfection and procedures. Review training records of individuals performing cleaning and disinfection.
	Check area pressure differentials, particularly with respect to the last sanitization.
	Evaluate mechanical equipment in the area as possible source of contamination.
	Review prior and subsequent monitoring results.
	Review sterilization cycle records.
Facility surfaces	Perform investigation for possible sources of contamination.
	Evaluate cleaning and disinfection practices; review preparation of disinfectants, cleaning records, and training records of individuals performing cleaning and disinfection.
	Review possible unusual events during manufacturing operation.
	Examine areas during operation.
	Review closed-circuit video (if applicable).
	Verify that environmental controls were not circumvented.
	Review risk of direct or indirect product contact with impacted surfaces.
	Review isolates for occurrence in other types of EM tests.
	Evaluate condition of the room (e.g., peeling paint or cracks in ceiling, walls, and floor).

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(Table 5.7-1 Continued)

System	Investigation Elements
High purity water system (WFI, clean steam, purified water)	Review upstream water treatment systems (e.g., carbon beds used to remove chlorine from municipal water systems).
	Examine bioburden data for other samples or sites in system—port contamination vs. system contamination.
	Examine endotoxin and water chemistry data for system.
	Review efficacy of sanitization procedure and schedule.
	Inspect system preventive maintenance records. Evaluate impact on product.
	Verify integrity of sample collection and use procedures.
	Inspect system for dead-legs, proper sloping, hose management, and sample-port design and location. Review data for generation and distribution system for potential trends. Determine whether system is functioning properly. Review data for flow rate, pressure, and vent filter integrity wherever used.
Personnel gowning (gowning and gloves)	Evaluate possible operator impact on product.
	Review EM data and sterility test data.
	Review preparation and expiry dates for disinfectants used on gloves.
	Identify all morphologically unique isolates (human vs. environmental).
	Interview operator for potential cause and retrain or requalify operator. Check the system for integrity of isolators and RABS gloves.
	Evaluate training of operator. Review sanitization and disinfection records of area.
	Review closed-circuit video (if available).
	Review previous gowning data for the operator and other operators on the same day.

The points listed in the table are not all-inclusive, as these recommendations are intended only to suggest investigative activities. Corrective actions based on process knowledge and understanding can be implemented when sampling and laboratory failures have been ruled out. Appropriate corrective actions arise holistically from evaluation of the investigation elements, leading to the root cause.

The reviewer may employ scientific judgment to postpone any corrective action until the result is confirmed and/or an investigation has been completed. It may also be appropriate to provide management with a routine summary of action-level excursions for review. All corrective actions listed include an evaluation of the action for effect on the product.

5.8 Documentation

The types of records that should be maintained as part of a routine EM program include:

1. Date and time of sampling
2. Product identification
3. Identification of the individual performing test
4. Test method or procedure reference
5. Activity level at site during test
6. Sampling equipment identification
7. Physical parameters like temperature, relative humidity, and differential pressurization
8. Sample site
9. Area classification
10. Schematics of areas showing sample site locations
11. Sample site criticality
12. Sampling frequency
13. Test results with units (e.g., CFU/plate/hour)
14. Identification of the analyst recording results

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15. Date results read
16. Alert level and/or action limit
17. Temperature and duration of incubation
18. Traceability of the certification date, release date, lot number, and expiration date of media used
19. Microbial Characterization of contaminants
20. Name of reviewer
21. Approval of data
22. Review of historical data
23. Calibration due date for sampling instrumentation and equipment
24. Methodology and analysis used to specify action and/or alert levels
25. System for documenting investigative and corrective action:
    - Description of deficiency
    - Possible causes of problem
    - Identification of persons responsible for relevant corrective action
    - Description of action steps and their schedule for implementation
    - Evaluation of effectiveness of corrective and preventative action steps

6.0 Environmental Monitoring by Manufacturing Application

The levels and types of monitoring conducted may vary based on the type of product being manufactured, the attributes of the product, and the manufacturing areas and processes used.

6.1 Terminal Sterilization

Environmental monitoring in support of terminal sterilization monitors microflora from all sources—manufacturing environment, process gases, water used in product formulations, sterilizer feed and cooling water—that have the potential to contribute to product bioburden or endotoxin. This includes testing of air, surfaces, and water for pharmaceutical use.

Microbial levels of incoming containers and closures may also be monitored periodically (e.g., if the product is not terminally sterilized using an overkill cycle). While control of the environment in which the products are prepared is important, the most critical aspect of the program is the bioburden of the product and components to be sterilized, including both the number and type of microorganisms found.

6.2 Aseptic Processing

Environmental monitoring in support of aseptic processing is specifically designed to determine the number and type of microorganisms associated with the direct assembly or preparation of product prior to sealing the filled containers. The number of sample sites and frequency of monitoring are considerably greater than those for established terminal sterilization programs. Air, water, personnel, compressed gases, machinery, and other surfaces within the filling room and associated support areas are routinely monitored. Microbial levels may be periodically checked on incoming containers and closures. Adequate environmental control is an integral part of the aseptic manufacturing process and is critical to sterility assurance. EM data must be reviewed prior to product batch release and investigation of excursions closed. EM should also be conducted when operator interventions are required during aseptic processing.

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6.2.1 Isolation Technology

The EM program for isolators differs from conventional aseptic filling operations and is based on the risks associated with isolator technology especially the physical integrity of half-suits (if applicable) and gloves. Monitoring of air, surfaces, and personnel outside of the isolator environment is not required after each filling operation; however, monitoring of isolator gloves and half-suits is required at the end of each filling operation or campaign (as applicable) to detect potential contamination. Product contact surfaces that are sterilized or changed out between batches in a campaign should be monitored at the end of the batch specific activities, prior to sterilization / change out.

6.3 Air

Effectively designed air-handling units and use of high-efficiency particulate air (HEPA) filters or ultralow particulate air filters bring air of appropriate quality to cleanrooms. Although the use of HEPA filters and air exchange rates to remove particles from the incoming air is a very effective way to reduce the particle load in an environment, especially under at-rest conditions, abnormal activity levels of equipment and people in a room require an increase in the air exchange rate for the area in order to remove the contaminants quickly. People are considered a major contributor of particulates, especially microorganisms, to the environment in aseptic areas. Air monitoring determines if there are viable and/or total airborne particulates in locations that would allow them to settle on product-contact surfaces and thereby find their way into process intermediates or final product. The analysis of trends and frequencies of viable and total airborne particulates is also vital to the understanding of the environmental impact of personnel behavior and process activities. Air-sampling methods and frequencies of monitoring will depend on manufacturing process needs and should be risk-based.

6.3.1 Total Airborne Particulate Monitoring

Monitoring of total airborne particulates is a necessary component of an EM program. It demonstrates control of potential contaminants in the environment to which the product is exposed during the manufacturing process. Cleanroom and other controlled areas are classified based on the level of total airborne particulates alone or air cleanliness. Classified areas must consistently meet these particulate levels (see **Section 3.0**).

ISO Standard 14644-1 describes the classification of air cleanliness for cleanrooms and clean zones based on specified concentrations of airborne particulates per volume of air sampled. It prescribes methods for verifying air cleanliness in specified particulate size ranges. The sample volume for routine monitoring does not need to be the same as that used for formal classification of cleanrooms ***(8)***.

For products marketed in the United States, the FDA requires that total airborne particulates of 0.5 µm or larger be measured as part of cleanroom classification and routine monitoring. ***(14)*** At the time of this Technical Report revision, requirements outside of the United States (e.g., EU and PIC/S Annex 1) also include measurements for particles of 5 µm or larger for both classification and monitoring purposes ***(10,11)***.

For aseptic processes, some regulatory guidance documents require monitoring in each production shift. Some regulators recommend remote counting systems and continuous monitoring. For remote systems, the distance from the sample collection port to the detection or quantification technology should be considered for the potential of particle loss. Design of the monitoring system should comply with the system manufacturer's recommendations and align with ISO 14644 in that the length of tubing should typically not exceed one meter and the radius should be greater than 15 cm. Continuous monitoring of particles is required for aseptic processing rooms of Grade A (ISO 5) classification during operation. Some operations can generate high levels of product particles (e.g., powder) that, by their nature, do not pose a risk of product contamination. In these cases, it is acceptable to monitor prior to and post-exposure to the risk since it would be impossible to distinguish extrinsic contamination to intrinsic particle generation. Additionally, continuous monitoring of environments with intrinsic particle generation may damage the particle

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measuring equipment. Initial qualification of the area under dynamic conditions without the actual filling function provides some baseline information on the non-product particle generation of the operation.

Generally, routine monitoring of total airborne particulates should be performed within a one-foot radius (approximately 0.3 m) of the identified critical location, if possible, and also at critical working or processing height. However, sampling should not be intrusive to the process.

6.3.2 Viable Particulates

Viable particulate determinations are critical in monitoring air quality in controlled environments. Traditional growth-based techniques for monitoring viable particulates in air are limited by:

- Type of equipment
- Growth media
- Incubation time
- Operator contamination of the capture media

6.3.2.1 Sampling Sites

The principles mentioned for site selection in **Section 5.2** apply here as well.

6.3.2.2 Methods

Regulatory agencies currently require active air sampling of environments on a routine basis to demonstrate control of viable airborne particulates. Some agencies require that passive sampling methods such as settling plates should be performed and justification provided if a company does not use passive sampling for select operations. Generally, active sampling methods are required with operating levels defined per unit volume of air. Both active and passive sampling methods may be used together to gain knowledge and understanding of changes and trends in the cleanroom environment. Active sampling methods have the benefit of sampling large homogeneous volumes of air in short amounts of time, but these methods may cause disruptions in the airflow of the environment being sampled. Therefore, these sampling methods should be included in air flow visualization studies in order to assess their potential impact on the aseptic environment. Alternatively, passive sampling methods can be used for prolonged periods of time (up to four hours) in close proximity to critical processes without affecting the airflow of the environment. An additional advantage of passive sampling is that it mimics the potential pathway of contamination to a critical process.

6.3.2.3 Equipment

Microbial air samplers are typically based on two principles —inertial or centrifugal impaction—to capture and deposit the organisms onto the agar or filter surface. No microbial air samplers are 100% efficient at detection or collection of all viable particles present. The physical efficiency of an air sampler is referred to as the "cutoff size" or "d_{50} value" (i.e., the size of a particle in microns that the sampler has a 50% chance of recovering), which is supplied by the equipment manufacturer. Some portable viable samplers are capable of remotely sampling room air as well as compressed gases with the use of adapters. Other samplers are dedicated for specific sampling types (portable, remote, isolator, RABS, compressed gas, etc.). Features of each sampler should be discussed with the individual manufacturer.

Requirements for the validation of equipment are the responsibility of the manufacturer or a third party designated by the manufacturer. Physical and biological efficiency data presented by the manufacturer of the device should be sufficient to implement the sampling device (see also ISO 14698-1) ***(25)***. However, equipment must be qualified; typical user requirements are:

- Calibration of the instrument
- Verification of volume accuracy

Alternative microbiological methods may require more extensive validation. For a more detailed discussion, refer to PDA *Technical Report No. 33 (Revised 2013): Evaluation, Validation and Implementation of Alternative and Rapid Microbiological Methods* ***(22)***.

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6.4 Surface Monitoring

In addition to conducting viable air and total airborne particulate monitoring to determine the microbial contaminants and particulates surrounding the manufacturing operations, surface monitoring is conducted to determine the bioburden of surfaces, e.g., walls, floors and ceilings, in the manufacturing area as well as nonproduct- and product-contact surfaces of equipment.

6.4.1 Monitoring Considerations

The method of testing should be considered when developing the sampling plan. When periodic surface monitoring is performed, it should be done after the filling operation or a campaign has been completed so no extraneous contamination or residual growth media are introduced during the filling activities. The basic methods—contact plates, flexible films, swabs, and surface rinses—each provide surface monitoring data, (qualitative or quantitative) that can be used to assess environmental quality. The accuracy, reliability, and reproducibility of the sampling is affected by the collection and handling of samples. Proper personnel training is essential to ensure consistent and standardized sample collection. The type of media used will determine the ability to detect the representative flora from the sample site. Neutralizers are typically added to the media to inactivate chemical disinfectant residues on surfaces. Additionally, validation should be performed on the test method to confirm that sanitizing agents do not influence the recovery of microorganisms.

Recovery levels of surface-monitoring methods are typically low, due to the variability and/or composition of surfaces, variability in sampling procedure, analytical methods being employed (i.e., dilution), and the use of growth-based techniques. However, consistent sampling techniques will generate data that can be trended and analyzed over periods of time. These trends and data provide key insights into the efficacy of cleaning and disinfectant programs and potential impact of manufacturing activities in the environment being assessed.

6.4.1.1 Contact Plates

Contact plates are easy to use and can provide quantitative results, i.e., CFU per surface area sample. The plates are filled with sufficient neutralizing media to provide a convex surface to support the growth of microorganisms. The contact plate is pressed against a flat surface by gently rolling the plate from front to back on the sample area. The sample plate is then placed in an incubator for the required period of time as determined during method validation. Colonies, if present, are counted at the end of the incubation period.

Prior to use, the type of media and incubation conditions must be qualified. Such qualification may include laboratory studies using the compendial growth promotion organisms and/or representative organisms recovered from the facility environment. The use of irradiated plates in ISO 5 areas is required and is recommended in ISO 7 and ISO 8 areas. Contact plates should be allowed to acclimate to room temperature before use to minimize confluent organism growth due to condensation.

Contact plates are not suitable for sampling irregular surfaces; swabs are recommended for that situation. The media residue contact plates leave behind must be removed from the sample site after sampling has been completed.

6.4.1.2 Flexible Films

A flexible film containing media can be used in a manner similar to a contact plate or to provide a defined sampling area. The surface of the media film is pressed against a flat surface in a rolling motion to ensure full contact of the flexible film. The film is then placed in the incubator for the required period of time determined during method validation. Colonies, if present, are counted at the end of the incubation period. Like the contact plate, flexible films leave a media residue that must be removed from the sample site post-sampling.

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6.4.1.3 Swabs

Swabs are employed for equipment and irregular surfaces that are not suitable for contact plates or flexible films. Swabbing may also be performed on flat surfaces provided a template of approximately 2 in by 2 in (~25 cm^2) is used to define the sample area (however, contact plates are recommended for flat surfaces).

Swabs are composed of materials such as Dacron® or nylon manufactured in a flocked or spun material format. Prior to plating, each swab is vortexed in an appropriate diluent (e.g., sodium chloride, phosphate buffer, or peptone solutions containing neutralizers if needed). Swabs may provide qualitative (e.g., immersing the swab directly into liquid growth medium) or quantitative (e.g., processing the swab to obtain individual colony data) results depending on how they are processed after sampling ***(26)***.

6.4.1.4 Surface Rinse Method

The surface rinse method is best used for when a large surface area sample is required (e.g., kettles, equipment, trains, tanks). A sterile diluent should be used as the rinsing fluid that comes in contact with the interior surface and is then collected and tested. Membrane filtration is used to test the rinse fluid, because of the large sample volume, and the applied holding time of the rinsing samples prior to plating should be qualified.

6.5 Personnel Considerations

Personnel are the primary source of contamination, which is why all cleanroom personnel (including maintenance workers and others who enter periodically when cleanrooms are in operation) must be carefully trained and qualified before entering an aseptic environment.

The qualification training should include:

- Attention to personal hygiene
- Basic microbiology education or background
- Aseptic techniques
- Appropriate cleanroom behavior
- Patient safety hazards posed by a contaminated product
- Gowning certification
- Participation in aseptic processing simulation (APS or media fill) activities, where applicable

After initial training, personnel should participate regularly in an ongoing training and monitoring program. Routine microbiological monitoring of personnel garments and gloves should also be completed to assess the ongoing practice of aseptic technique. Supervisory personnel should routinely evaluate each operator's conformance to written procedures during actual operations. Likewise, the quality unit should provide regular oversight to ensure that established, written procedures and aseptic techniques are being followed during manufacturing operations.

6.5.1 Training and Certification of Personnel for Aseptic Manufacturing Areas

All training and certification activities should be documented and kept as part of the employee file.

6.5.2 Causes for Requalification or Retraining

Companies should have a policy stating when requalification or retraining is necessary, for example:

- Retraining on general aseptic practices and techniques should be provided at least annually
- Recertification should be required after extended absences (e.g., for illness or family leave)
- Recertification and/or retraining should be considered as corrective action for trends in personnel monitoring excursions

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6.6 Environmental Monitoring During Product Sterility Testing

Sterility-testing facilities should be monitored to demonstrate the absence microbial contamination ***(4,10,14)***. Routine types of monitoring that are conducted are as follows:

- Viable air monitoring
- Surface monitoring (including isolator gloves if applicable)
- Personnel monitoring

Total airborne particulate monitoring is not mandatory; however, it may provide useful information regarding the HEPA filters in the ISO 5 testing environment or the integrity of isolators. Lab materials may generate particles that are not representative of the actual manufacturing conditions, and there is no risk to marketed product in the testing environment.

6.7 Environmental Monitoring in Grade A Air Supply

The term "Grade A air supply" is introduced in Annex 1 and is specifically used to support the capping of fully stoppered vials until the cap has been crimped by providing additional protection without the requirement to meet all criteria related to a Grade A area (e.g., environmental continuous total particulate monitoring or environmental viable monitoring limits).

Grade A air supply provides air which is passed through a filter qualified as capable of producing Grade A total particulate quality air, but where there is no requirement to perform continuous total particulate monitoring or meet Grade A viable monitoring limits and the area itself is not classified.

As Grade A Air Supply is not a clean room category (e.g., according to ISO 14644), no guidelines for viable and total particulate limits exists. In order to confirm Grade A total particulate quality air is supplied, testing at the point of supply of the filtered air should be performed as a minimum during qualification and requalification.

At rest viable (air and surfaces) and total particulate monitoring may be performed based upon a risk assessment. Routine monitoring, limits, and positioning should reflect the requirements for this "clean process" and be aligned with the contamination control strategy and meet criteria applied for the background of the Grade A Air supply (usually Grade C, but minimum Grade D).

6.8 Water

As water is used as a raw material in the production, processing, and formulation of products, this document only addresses the process monitoring of the water for microbial attributes. Most countries have compendial requirements for evaluation of water quality The recently published EMA Guideline for the Quality of Water for Pharmaceutical Use should be consulted as to the type of pharmaceutical-grade water used manufacturing ***(27–30)***.

The type of water selected for manufacturing is based on the type of product and process being used. The water quality should be evaluated for microbial, bacterial endotoxin, and chemical attributes in accordance with the type of water and applicable compendial or regulatory requirements. Water quality is critical for the manufacturing of parenteral drugs and selected devices because it is used in product formulation, product-contact component and equipment washing, and final rinsing of equipment and devices. When purified water is used to manufacture other types of products or intermediates (e.g., solid dosage forms, API, small molecules), the control of microbial quality is critical to minimize the buildup and spread of microbial contaminants in the facility as well as on manufacturing equipment. The quality of the laboratory water should meet the needs of analytical methods and technologies, which may require compendial-quality laboratory water. **Table 6.8-1** lists the pharmacopeial water GMP specifications used for manufacturing water and distribution system for pharmaceutical use.

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Table 6.8-1 Examples of measurements critical to water for Injection manufacturing and distribution process

Critical Parameters Measurement Quality Attributes	US Pharmacopeia (USP) 2021	European Pharmacopeia (EP) 9:8 2019, and 10,	Japanese Pharmacopeia (JP) 17, 2016	Chinese Pharmacopeia (ChP) 2015
Source water	Drinking water source, Purified water	Suitable for Human Consumption	JP Drinking water specification or purified water	Purified water
Purification Process	Distillation or equivalent/superior process			Distillation
pH	N/A			5.0-7.0
Conductivity at 25°C	1.3 µS/cm In-line Stage 1 (Stage 2-3 offline)[1,2]			
WFI Microbial Limit Specification	NMT 10 CFU/100 mL			
Maximum TOC	500 ppb or 0.5 mg/L[3]			
Bacterial Endotoxin Limit	< 0.25 (EU/mL)			
Nitrates Limit (ppm)	NA	0.2	NA	0.06
Nitrites Limit (ppm)	N/A			0.02
Ammonia Limit (ppm)	N/A			0.2
Heavy metals Limit (ppm)	N/A			0.1

[1] In-line/stage 1 conductivity specification at 25°C with other values at other temperatures. For usual WFI, the USP additionally has off-line stage 2 and 3 specifications which may alternatively be met.

[2] Performed as an atmosphere and temperature equilibrium test. Alternatively, for JP Chapter 21, the test may be performed in-line with specified limit at 25°C and other values at other temperatures. There is no separate Stage 3-like option in JP.

[3] There is no specification for in-line TOC test USP 2021 Issue 1, JP recommends Recommended levels of TOC ≤ 300ppb (in-line) ≤4 00ppb (off-line)

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6.8.1 Sample Site Selection and Frequency of Monitoring

The sampling and frequency of monitoring for water systems should be sufficient to ensure that the water quality meets its intended use specifications and appropriate regulatory requirements.

The frequency of monitoring and sampling site selection depends on the type of water and manufacturing process, system configuration, and facility design. The frequency of sampling should be designed to facilitate detection and analysis of trends in chemical attributes, microbial flora, and bacterial endotoxin.

High-purity water systems should be validated to demonstrate that design requirements are consistently met and maintain a state of control. The site selection and frequency determination should be justified by thorough review of the water system validation performed and, using risk-based approaches, by type of product and manufacturing processes in concert with pharmacopeia and regulatory requirements. See **Section 10.0** (**Appendix 2)** for suggested sites and sampling frequencies.

Once a water system is validated to be in a state of control, appropriate samples should be taken from the water source or generation system, holding and storage tanks, and distribution system to assess the microbiological quality for its intended use. Initial testing of water systems typically includes sampling after each process step. The testing of high-purity water systems should include the sampling of all point-of-use sites.

6.8.2 Sample Collection and Testing

Water samples should be collected in a manner that is consistent with manufacturing practices. For example, if manufacturing flushes use points prior to use, samples should be collected with the same flush cycle. On the other hand, if manufacturing does not flush use points, there should be no flush prior to sample collection. All sample sites should be in a good state of repair and fully functional to support the monitoring program (including consideration for the use, maintenance, and proper drainage of flexible hoses). Distribution-system sample locations aside from the point-of-use sites should be chosen carefully to demonstrate microbiological quality throughout the distribution system. Additionally, it is recommended to not use IPA on gloves or sample port prior to TOC sampling to not generate false positives.

Microbiological examination of water should be initiated as soon as possible after collection of the sample. If immediate processing is not possible, samples should be refrigerated at 2–8 °C. The time elapsed between collection and examination should be justified in terms of microbial survival or stability, including considerations for storage and shipping to third-party testing facilities. **Table 6.8.2-1** provides and overview of current guidelines for typical pharmaceutical water and clean steam distribution systems monitoring, methods, requirements, frequencies, and levels.

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Table 6.8.2-1 Typical Pharmaceutical Water and Clean Steam Distribution System Guidelines

	Clean Steam	Purified Water	USP Water for Injection	EP Water for Injection	JP Water for Injection	ChP Water for Injection
Site	Sample at the point of generation and the furthest point of use	Representative use points in the distribution system	All accessible use ports in the distribution system.			
Methods	Chemistry and LAL	Chemistry, pour plate (1.0 mL) or membrane filter (100 mL)	Chemistry, LAL, and membrane filter (200 mL)		Chemistry, LAL, and membrane filtration (100mL)	Chemistry, LAL, and membrane filtration (not less than 100 mL)
Microbiological test and incubation requirements (10 CFU/100 mL limit)	Not recommended	Plate Count Agar (PCA/TGYA)[1] for 48 – 72 hours[2] at 30 – 35 °C[3]		R2A agar, incubated at 30 – 35 °C for not less than 5 days	Standard agar media, incubated at 30 – 35 °C for 48 – 72 hours, or R2A agar, incubated at 20 – 25 °C or 30 – 35 °C for 4 – 7 days	R2A agar, incubated at 30 – 35 °C for not less than 5 days

[1] For optimum recovery, an alternative medium may be more appropriate (e.g., m-HPC, TSA/SCDA, R2A).

[2] For optimum recovery, alternative incubation times may be needed.

[3] For optimum recovery, alternative incubation temperatures may be needed.

6.9 Compressed Gas Monitoring

The use of compressed air and compressed gas in the aseptic ISO 5 environment may adversely affect the environmental conditions if appropriate precautions, routine testing, and critical assurances are not developed into a system. This section offers some points that should be considered.

Using a 0.2 µm filter on the compressed gas line prior to entry into the cleanroom is common practice. Integrity testing of the filter and sanitization of the line post-filtration may reduce the need for sampling. The sampling equipment must be compatible for use with compressed gases. A pre-filter may be installed in the system to reduce the bioburden to the point-of-use filter that will extend its life and reduce the possibility of higher contamination reaching the area.

The particulate or bioburden level of any gas or air expelled into the environment should be equal to or better than the classification level of the room ***(10)***.

7.0 Environmental Monitoring Prerequisites

This section briefly describes the scope of prerequisites for EM, which includes the following:

1. Good facility and equipment design
2. Media vendor audit and qualification
3. Selection of microbiological culture media for EM testing
4. Culture media release
5. Selection of microorganisms for growth promotion
6. Microbiological culture maintenance
7. Culture media shelf-life confirmation
8. Sample holding studies and transport

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9. EM sample incubation strategy and validation
10. PQ of all critical room parameters (e.g., temperature, air velocity, differential pressure, humidity)
11. Disinfection of the area and baseline sampling
12. EM monitoring equipment qualification
13. Analyst training and qualification
14. IQ/OQ of HVAC/Building automation system
15. Certification of HEPA filters
16. Airflow pattern measurements (at-rest and in-operation)

Note: Items 14-16 are fundamental prequalification activities that do not require further elaboration, so not discussed further in this section.

7.1 Facility Design

As stated in regulations, separate or defined areas of operation in an aseptic processing facility should be appropriately controlled to attain different degrees of air quality depending on the nature of the operation. Design of a given area involves satisfying microbiological and particle criteria as defined by the equipment, components, and products exposed, as well as the operational activities conducted in the area. Facility design is a crucial component of a good environmental control. Documents like ISPE's *Baseline Guide, Vol. 3: Sterile Product Manufacturing Facilities* describe the design, construction, commissioning, and qualification of sterile manufacturing facilities ***(31,32)***. Similar baseline guides are available for other types of product manufacturing.

The two-part ISO 14698 series (superseded by BS EN 17141:2020) provides extensive guidance on contamination control, and ISO 14644-1 provides methods to assess the control of air, surfaces, and liquid contamination ***(8,25,33)***.

7.2 Selection of Microbiological Culture Media for EM Testing

Microbiological growth medium, such as soybean-casein digest medium (SCDM) or trypticase soy agar (TSA) is suitable as it supports growth of a wide range of bacteria, yeast, and molds. This medium can be supplemented with neutralizing agents to overcome or to minimize the effects of sanitizing agents or antibiotics, especially for surface-monitoring purposes. Use of anaerobic growth media (e.g., TSA under anaerobic conditions) should be considered only in special circumstances. A general mycological, such as Sabouraud Dextrose or a modified Sabouraud Dextrose agar formulation, can be used as necessary when fungal contamination of the facility is suspected, although SCDM will adequately support growth of both yeast and mold ***(34)***. The media selected should be demonstrated to promote growth of Gram-positive and Gram-negative bacteria, yeast, and mold. Anaerobic agar media can be used for the recovery of microaerophilic and anaerobic organisms as needed based upon risk assessments, prior recovery of anaerobes, or sterility and/or APS failure as a result of anaerobes. Tryptic soy broth is typically used as the growth medium for swabs performed in Grade A environments (i.e., qualitative results). Alternatively, organisms can be recovered from swabs on suitable solid media (i.e., quantitative results). Culture media storage conditions and expiration dating should follow vendor recommendations. Each media lot should undergo release testing according to relevant pharmacopeial requirements including such quality control testing as appearance, growth promotion, and sterility.

Performing initial media qualification

- Validate the media to demonstrate that the fertility is not affected by desiccation at the end of the maximum allowed exposure period
- Where media require neutralizers, insure they are suitable for the applied disinfectants/agent.
- Ensure its suitability for use over the supplier's shelf life
- Perform growth promotion testing of media using low levels of microorganisms isolated from the manufacturing environment and obtained from recognized or national culture collections.

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- Document the use of microorganisms isolated from the manufacturing environment in an approved rationale.

Performing routine media testing

- Justify and document the quality control strategy for media used for environmental monitoring, including the need to perform sterility checks and negative controls.
- Perform growth promotion testing on each delivery of a given batch of media. Consider reduced testing based on risk assessment, including appropriate supplier control confidence and supplier quality agreement.
- Perform growth promotion testing of medium using micro-organisms as per requirement above

7.3 Selection of Microorganisms for Growth Promotion

Selection of microorganisms for growth promotion should be established according to relevant pharmacopeial guidance, and with consideration of USP <61>, based on the media to be evaluated ***(35)***. Microorganisms selected should include those required by the appropriate compendial chapter and representative environmental isolates. Selection of the environmental isolates should be based on microbial profiling assessment using the data from the framework of routine trending reports. For a new facility, organisms initially recovered from cleanroom qualification activities can be used. A microbiological culture maintenance program should be established that includes a procedure for testing and release with appropriate documentation of used passages (see USP <1117>) ***(36)***.

7.4 Sample Holding Studies and Transport

EM samples are sensitive to handling and storage conditions, which are deemed critical parameters. These conditions are affected by time of sampling and temperature of storage. Sample hold studies should be performed to validate the sample handling from end of sampling through initial incubation. Care should be used if transferring EM media using sanitizing airlocks (e.g., through sanitizing airlocks equipped with UV lights).

7.5 Environmental Monitoring Sample Incubation Strategy and Validation

Guidance documents and practices within the industry differ regarding incubation conditions. Information is available regarding selection of a suitable microbiological growth medium, but information about appropriate incubation times and optimum temperature conditions is limited or sometimes conflicting.

Typically, the total aerobic bacterial count can be obtained by incubating at 30–35 °C for 48–72 hours. Total combined yeast and mold count can generally be obtained by incubating at 20–25 °C for 5–7 days. Longer incubation times may be considered when contaminants are known to be slow growing. Industry surveys have shown a majority of companies using a two-temperature incubation with the lower temperature first.

Multiple incubation schemes can be employed if scientifically justified and validated to provide an appropriate level of sensitivity. These are typically laboratory-based in-vitro studies using cultured microorganisms, a combination of reference cultures such as ATCC strains and in-house isolates recovered from the production environment. Studies, such as those published by Gordon, et.al., demonstrate that there may be a difference between in vitro and in situ testing ***(37)***.

7.6 Environmental Monitoring Equipment Qualification

Assessment may be required to determine gaps in equipment qualification. The following items should be considered as prerequisite:

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- Equipment and instrument listing
- Documentation requirements
- Maintenance and calibration (e.g., air volume accuracy requirements)
- Tube length requirement for total airborne particulate monitoring equipment
- Data collection, transmission and storage, alarm conditions
- Installation and Operational qualification of EM equipment

7.7 Microbiology Analyst or Operator Training and Qualification

Appropriate training curricula and qualification requirements should be established for EM analysts or operators. That entails relevant, cohesive standard operating procedures, including gowning and gowning qualification for aseptic areas, and a good understanding of the procedures, their importance in aseptic manufacturing operations, and the impact or risk to quality for not adhering to these procedures.

8.0 References

1. Parenteral Drug Association. *Technical Report No. 69: Bioburden and Biofilm Management in Pharmaceutical Manufacturing Operations*; PDA: Bethesda, Md., 2015.

2. U.S. Pharmacopeial Convention. General Chapter <1072> Disinfectants and Antiseptics. In *USP 43–NF 38*, USP: Rockville, Md., 2004; p 7608.

3. Parenteral Drug Association. *Technical Report No. 30 (Revised 2012): Parametric Release of Pharmaceutical and Medical Device Products Terminally Sterilized by Moist Heat*; PDA: Bethesda, Md., 2012.

4. Parenteral Drug Association. *Technical Report No. 62: Recommended Practices for Manual Aseptic Processes*; PDA: Bethesda, Md., 2013.

5. Parenteral Drug Association. *Technical Report No. 44: Quality Risk Management for Aseptic Processes*; Bethesda, Md., 2008.

6. International Council for Harmonisation. *Quality Guideline Q9: Quality Risk Management*; ICH: Geneva, 2005.

7. International Organization for Standardization. *ISO 14937:2009 Sterilization of Health Care Products — General Requirements for characterization of a Sterilizing Agent and the Development, Validation and Routine Control of a Sterilization Process for Medical Devices*; ISO: Geneva, 2009.

8. International Organization for Standardization. *ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of air cleanliness by particle concentration*; ISO: Geneva, 2015.

9. U.S. Pharmacopeial Convention. General Chapter <1116> Microbiological Control and Monitoring of Aseptic Processing Environments. In *USP 42–NF 37*, USP: Rockville, Md., 2019; Vol. USP 42–NF 37, p 7701.

10. European Commission. *Annex 1: Manufacture of Sterile Medicinal Products (corrected), EudraLex – Volume 4 – EU Guidelines to Good Manufacturing Practice, Medicinal Products for Human and Veterinary Use* European Commission: Brussels, 2008(rev).

11. Pharmaceutical Inspection Convention/ Cooperation Scheme (PIC/S). *Annex 1: Manufacture of Sterile Medicinal Products,*

 © 2022 Parenteral Drug Association, Inc.

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Guide to GMP for Medicinal Products; PIC/S: Geneva, 2018.

12. World Health Organization. *Annex 4: WHO Good Manufacturing Practices for Sterile Pharmaceutical Products, WHO Technical Report Series 957*; WHO: Geneva, 2010.

13. Ministry of Health, Labour and Welfare of Japan. Microbiological Examination of Non-sterile Products–4.05: Microbial Examination of Non-sterile Products. In *The Japanese Pharmacopoeia, 17th Edition (English version)*, MHLW: Tokyo, 2004; pp 2486-87.

14. U.S. Food and Drug Administration. *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing—Current Good Manufacturing Practice*; U.S. Department of Health and Human Services: Rockville, Md., 2004.

15. Parenteral Drug Association. *Technical Report No. 70: Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*; PDA: Bethesda, Md., 2015.

16. Parenteral Drug Association. *Technical Report No. 54-4: Implementation of Quality Risk Management for Pharmaceutical and Biotechnology Manufacturing Operations, Annex 3: Case Studies in the Manufacturing of Biotechnological Bulk Drug Substances*; PDA: Bethesda, Md., 2014.

17. Sandle, T. Environmental monitoring risk assessment. *Journal of GXP Compliance* **2006,** *10* (2), 54-75.

18. Biophorum Operations Group (BPOG). *Environmental Monitoring (EM): Harmonized Risk-Based Approach to Selecting Monitoring Points and Defining Monitoring Plans*; BPOG: London, 2019.

19. Parenteral Drug Association. *Elements of a Code of Conduct for Data Integrity in the Pharmaceutical Industry*; PDA. PDA: Bethesda, Md., 2016. https://www.pda.org/scientific-and-regulatory-affairs/regulatory-resources/code-of-conduct/elements-of-a-code-of-conduct-for-data-integrity-in-the-pharmaceutical-industry (accessed Aug 15 2019).

20. Parenteral Drug Association. *Technical Report No. 80: Data Integrity Management System for Pharmaceutical Laboratories*; PDA: Bethesda, Md., 2018.

21. Parenteral Drug Association. *Technical Report No. 59: Utilization of Statistical Methods for Production Monitoring*; PDA: Bethesda, Md., 2012.

22. Parenteral Drug Association. *Technical Report No. 33 (Revised 2013): Evaluation, Validation and Implementation of Alternative and Rapid Microbiological Methods*; PDA: Bethesda, Md., 2013.

23. Council of Europe. Alternative Methods for Control of Microbiological Quality, Chapter 5.1.6. In *European Pharmacopoeia 10.0, 10th Edition*, Council of Europe: Strasbourg, 2019; p 628.

24. U.S. Pharmacopeial Convention. General Chapter <1058> Analytical Instrument Qualification. USP 42-NF 38: Rockville, Md., 2017; p p. 7522.

25. International Organization for Standardization. *ISO 14698-1:2003 Cleanrooms and Associated Controlled Environments — Biocontamination Control — Part 1: General Principles and Methods*; ISO: Geneva, 2003.

26. Dalmaso, G, et al. Qualification of High-Recovery, Flocked Swabs as Compared to Traditional Rayon Swabs for Microbiological Environmental Monitoring of Surfaces. *PDA J Pharm Sci and Technol* **2008,** *62* (3), 191-99.

27. Council of Europe. Total Organic Carbon in Water for Pharmaceutical Use, Chapter 2.2.44. In *European Pharmacopoeia*, Council of Europe: Strasbourg, France, 2008; Vol. 6th Edition.

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28. Ministry of Health, L a W o J. General Information–G8 Water: Quality Control of Water for Pharmaceutical Use. In *The Japanese Pharmacopoeia, 17th Edition (English Version)*, MHLW: Tokyo, 2016; p 2543.

29. U.S. Pharmacopeial Convention. *General Chapter <643> Total Organic Carbon*; USP: Rockville, Md., 2012; p 6795.

30. U.S. Pharmacopeial Convention. General Chapter <1231> Water for Pharmaceutical Purposes. In *USP 42–NF 37*, USP: Rockville, Md., 2016; p 8121.

31. Gaison, E. Controlling Microbiological Contamination: Standards, Techniques, and a Case Study. *Med Dev Diag Ind* **1998,** *10* (2), 34–39, 66.

32. International Society for Pharmaceutical Engineering. *Baseline Guide, Volume 3: Sterile Product Manufacturing Facilities, Third Edition*; ISPE: Bethesda, Md., 2018.

33. International Organization for Standardization. *ISO 14698-2:2003/COR 1:2004 Cleanrooms and Associated Controlled Environments — Biocontamination Control — Part 2: Evaluation and Interpretation of Biocontamination Data — Technical Corrigendum 1*; ISO: Geneva, 2004.

34. Marshall, V, et al. Comparative Mold and Yeast Recovery Analysis (The Effect of Differing Incubation Temperature Ranges and Growth Media). *PDA J Pharm Sci and Technol* **1998,** *52* (4), 165-69.

35. U.S. Pharmacopeial Convention. General Chapter <61> Microbiological Examination of Nonsterile Products: Microbial Enumeration Tests. In *USP 43–NF 38*, USP: Rockville, Md., 2013; p 6457.

36. U.S. Pharmacopeial Convention. General Chapter <1117> Microbiological Best Laboratory Practices. In *USP 39–NF 34*, USP: Rockville, Md., 2016.

37. Gordon, O, et al. Comparison of Different Incubation Conditions for Microbiological Environmental Monitoring. *PDA J Pharm Sci and Technol* **2014,** *68* (5), 394-406.

9.0 Appendix 1: Qualification of Cleanrooms (Classified Areas) Used for Manufacturing and Filling of Sterile Products

9.1 Purpose

This appendix provides a consistent methodology for qualification of cleanrooms (classified areas) as defined in this technical report and used for the manufacture, processing, packaging, holding, and testing of active pharmaceutical ingredients, bulk, intermediates, and finished dosage forms of sterile drug products, biologics, and medical devices for human use.

European Union and World Health Organization (WHO) designations for cleanroom classifications are used throughout, as these are widely used and specific to their application as described in **Section 4.0**. Applicable ISO classifications are added where appropriate.

9.2 Scope

Section 9.0 (Appendix 1) focuses on the activities necessary to provide a high degree of assurance that cleanrooms used meet the requirements necessary to ensure product quality. Provided that appropriate environmental monitoring (EM) and/or alarm systems are in place, HVAC mechanical and

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control systems have an indirect impact on product quality. These HVAC systems should be commissioned to ensure that they are suitable for their intended purpose and have been installed according to applicable engineering standards and specifications. Formal qualification activities (i.e., following standards and practices dictated by good manufacturing practices (GMP)) should center on final HEPA filtration, including ductwork beyond the final filter, and on monitoring of area parameters to ensure that the cleanroom requirements are met. Applicable tests for these activities are described in **Section 4.0**.

Areas designated as controlled (non-classified) are sometimes employed as peripheral areas of sterile product manufacturing facilities. (This class of cleanroom may also be known by other names, e.g., clean non-classified (CNC) and pharmaceutically controlled area (PCCA).) These areas do not involve exposure of materials or components that will be sterile and are separated from critical zones by several cleanliness levels. Qualification of these areas may not be required but ensuring that their design criteria have been met and they are fit for the purpose specified is recommended.

Isolators and unidirectional (laminar) airflow units that provide Grade A air to local areas within a larger cleanroom are included in the scope. Other locally controlled environments (e.g., freezers, refrigerators, stability chambers) that are self-contained and where the environment is provided by dedicated specialized equipment rather than by traditional HVAC systems are excluded. While sterilization tunnels or other devices that use unidirectional or HEPA-filtered air are not covered by **Section 9.0** (Appendix 1), the test methods described for the airflow, filter testing, and air classification may be directly applied.

9.3 Procedure

At a minimum, the following inspections and tests should be considered and applied based on risk to product quality. In general, these are recommended for all cleanrooms described in this technical report.

9.3.1 Installation Qualification (IQ)

Installation qualification (IQ) verifies that critical system components meet user requirements and have been properly identified, installed, and calibrated (instruments). Critical components are identified by component risk analysis as those that have a direct effect on product quality. IQ requirements may be confirmed through inspection or by referencing commissioning or factory acceptance test documentation, including:

Items to address during IQ include:

- Description, location, and use of the cleanrooms/classified areas to be qualified
- Verification of the type, range, and placement of sensors and alarms measuring, indicating, and/or reporting environmental parameters having a direct effect on product quality (see Operational Qualification also)
- Verification of the location of final system filters where the environmental classification has airborne viable and/or total airborne particulate limitations (e.g., Grades A–D)
- Verification of material and proper installation of ductwork downstream of the final HEPA filter
- Completion of a commissioning plan or document, including a test and balance report, approved by the system user and responsible engineering group
- Description, location, and use of the classified area including the defined boundaries and all relevant cleanrooms
- Verification of critical system components, including terminal air filters, instruments, and alarms identifying manufacturer, model number, size/range/efficiency (as appropriate), materials (product contact only, not typical for HVAC) and, where appropriate, material certification of components. (In cases where a room monitoring system (RMS) is employed that includes the

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monitoring sensors, recording, and alarm functionality, those components may be documented in a separate protocol).
- Verification of instrument calibration certifications in compliance with site calibration procedures
- Software or firmware title/identification, version, and installation verification (may be in separate RMS protocol)
- Transfer of equipment, instrument, and information on maintenance, repair, and operating materials to the appropriate site maintenance records (e.g., computerized maintenance management system
- Confirmation that maintenance procedures and/or manuals are in place

9.3.2 Operational Qualification (OQ)

Operational qualification (OQ) verifies operational requirements that are critical to quality. These are limited to the environmental parameters to which product may be exposed (e.g., temperature) and the operation of final filters required to control airborne contamination. Operational characteristics having indirect impact on the environment, including air handler operation (e.g., fan speed, pre-filters, coil performance) and control elements (e.g., variable air volume controls) should be included in commissioning.

Items to address during OQ include:
- Final air filter pressure differential
- Filter leak test (for H13 and H14 filters or better only)
- Air flow velocity (if unidirectional flow is required)
- Air flow visualization (smoke) study
- Room pressure differential
- Temperature
- Relative humidity
- Alarm and interlock checks
- Room air classification (total airborne particulate count)
- Air exchange rate
- Recovery time (correlated to air exchange rate)

OQ is to be performed at rest, with HVAC equipment installed, commissioned, and balanced. Other equipment in the room must be installed and functioning without product or material to the extent that is safe. Operating staff, except for require test personnel, are not present in the room. The operation of equipment verifies the ability of the system to control airborne particulate generated by the equipment.

Instruments used to gather OQ data may be installed permanently in the system or may be test instruments. They must be calibrated according to the site calibration standard operating procedures and must be identified in the OQ documents. OQ requirements may be confirmed by referencing the commissioning or factory acceptance test (FAT) documentation if the site practices for GMP documentation have been adhered to during those stages and the appropriate test methodology has been employed.

Prior to starting OQ, all relevant procedures, such as gowning and equipment operation, should be drafted and subject to interim approval for purposes applicable to OQ testing. (These may be modified prior to final approval for PQ.)

Unless otherwise noted, all testing in support of OQ should comply with the most current version of ISO 14644-1 *Cleanrooms and Associated Controlled Environments — Part 1: Classification of air cleanliness by particle concentration*. ISO 14644-3 *Cleanrooms and Associated Controlled Environments — Part 3: Test Methods* outlines the methodology to be followed for each test and provides acceptance criteria ***(1,2)***.

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The test matrix in **Table 9.3.2-1** outlines requirements for qualification in various cleanliness zones. The tests are not required to be performed in any prescribed order. Performing certain tests in parallel (e.g., room pressure differential, temperature, and humidity) is common practice.

Note: All tables in appendices indicate values for start-up sites, not for facilities where historical data is available.

Table 9.3.2-1 OQ Testing Requirements (At Rest)

Test	Grade A[3] (ISO 5 unidirectional)	Grade B (ISO 5)	Grade C (ISO 7)	Grade D (ISO 8)
Filter Pressure differential	Yes	Yes	Yes	Yes
Filter Leak Test[1]	Yes	Yes	Yes	Yes
Air Flow Distribution	N/A	Yes	Yes	Yes
Air Flow Velocity	Yes	N/A	N/A	N/A
Air Flow Visualization (smoke) Study	Yes	Yes	Optional[2]	Optional[2]
Differential room pressure	N/A	Yes	Yes	Yes
Temperature	Yes	Yes	Yes	Yes
Relative Humidity	Yes	Yes	Yes	Yes
Alarms and interlocks	Yes	Yes	Yes	Yes
Room Air Classification (Total airborne Particulate Count)	Yes	Yes	Yes	Yes
Recovery time	N/A	Yes	Yes	Yes

1 Required for H13/H14 or better filters only

2 Air flow visualization studies in non-unidirectional areas are optional based on risk analysis. See Section 9.6 below

3 OQ testing for Grade A air supplied for crimping of aseptically filled vials or for local assurance of particulate control (i.e., local unidirectional air flow devices) is the same as for Grade A, except for Unidirectional

The tests listed in **Table 9.3.2-1** are all generally accepted for initial qualification. OQ following a change is determined by a risk analysis depending on the type of change, including area and equipment configuration and change of conditions. For example, a change of HEPA filter does not impact temperature, humidity, or room differential pressure. Therefore, the recommended tests for filter changes are limited to air velocity (Grade A), air flow distribution (Grade B, C, D), filter differential pressure (all Grades), filter leak test (of H13/H14 or better filters), and room classification.

With the exception of the filter change described here, changes to equipment in critical areas (e.g., modifications or changes of the geometrical shape on an aseptic filling line in Grade A) require air flow visualization (smoke) studies.

9.3.3 Routine Recertification

Requalification or recertification of cleanrooms may be driven by change (see **Section 4.0**). Where changes have not driven recertification of an area, providing evidence that the acceptance criteria indicated in the initial qualification are still met is recommended. This may be performed according to the schedule in **Table 9.3.3-1** and its notes. It is important to regularly review results and trends of parameters that are continuously monitored or regularly recorded (e.g., temperature, humidity, room pressure differential, others based on risk analysis) and that frequencies are justified.

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Table 9.3.3-1 Routine Area Recertification[1]

Test[2]	Grade A	Grade B	Grade C	Grade D
Filter Leak Test[3]	6 mos.	6 mos.	Annual	Annual
Air Flow Distribution	N/A	Annual	Annual	Annual
Air Flow Velocity	6 mos.	N/A	N/A	N/A
Room classification	6 mos.[4]	Annual	Annual	Annual
Recovery[5]	N/A	N/A	N/A	N/A

1 All re-certification testing is performed "at rest"

2 For all areas, the frequency of testing maybe extended to twice the initial frequency after three successive successful tests, i.e., Grade B may be done annually, and other Grades may be tested every second year. Installation of new filters requires the initial frequency for a minimum of three successive successful tests.

3 Filter leak test is required only for H13/H14 filters or better.

4 For Grade A where continuous fixed particulate monitoring is in place re-certification of area classification is not required provided that the particulate monitoring satisfies the requirements of ISO 14644-1.

5 Recovery testing for routine re-certification is not required except in cases where the Air Velocity test has shown a significant reduction (>20%) in the number of air changes per hour (ACH). For changes in HVAC design, room layout or equipment configuration, Recovery testing must be considered within the risk analysis associated with the change.

During EM PQ and PQ of physical parameters, data is to be generated during in-operation conditions (i.e., with personnel present, equipment in place, routine cleaning and disinfection, and operations ongoing).

References

1. International Organization for Standardization. ISO 14644-1:2015 Cleanrooms and Associated Controlled Environments — Part 1: Classification of air cleanliness by particle concentration; ISO: Geneva, 2015.
2. International Organization for Standardization. ISO 14644-3:2019 Cleanrooms and Associated Controlled Environments — Part 3: Test Methods; ISO: Geneva, 2019.

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10.0 Appendix 2: Risk Assessment Examples

Table 10.0-1 shows a real-life example provided by Gapp Associates of a review of an existing EM program at a closed RABS filling operation (vial filling; Grade B background). Note that only 5 risk items - from a total of 19 - are listed in the table below.

Table 10.0-1 Example: Failure Mode and Effects Analysis (FMEA)

Risk Item	Problem Statement / Requirement	Risk Description / Potential Failure Mode and Impact	Causes of process/ product failure	Current Controls and Preventive Actions/ Comments SEV: assess the risk and impact on sterility and regulatory compliance, in case of a deviation/ deficiency to this statement/ requirement OCC: assess the probability, that the related locations and areas are - in fact- microbiologically contaminated DET: assess the probability, that the implemented EM sampling plan/ frequency/ number of samples/ methods would detect - in fact- a potential contamination	SEV	OCC	DET	RPN	Risk Class	Risk Accepted	Mitigation Measures/ CAPAs
1	"High-risk areas" in Grade A are monitored: samples are selected which are: • in close proximity to open product and critical surfaces • linked with an increased number of activities and interventions • linked with an extended time of critical interventions • linked with material flow into Grade A. These areas must be properly monitored and sampled in the EM program.	Potential contamination is not detected in high-risk areas, which results in product contamination in Grade A.	Ingress of particles and microbiological contamination into the high-risk areas by many activities and interventions/ extended activities/ material flow.	The high-risk areas in the RABS are a) the filling and stoppering area b) the stopper bowl and c) the turntable, where depyrogenated, open vials are exposed for an extended time. The above areas are well addressed by the surface monitoring program, but not by an active air monitoring/ surface monitoring at the stopper bowl. A settle plate is exposed. This is a deficiency, since high-risk interventions and aseptic activities are performed in the vicinity of the stopper bowl. (**Note:** active air monitoring is done during the set-up and once in a shift). **SEV: 3** (high impact on product sterility, in case that indirect product contact surfaces = stopper bowl is contaminated) **OCC: 2** (there are several interventions performed, therefore there is a moderate risk) **DET: 2** (no active air monitoring, linked with interventions, therefore reduced detectability and moderate risk; no surface monitoring of critical surfaces, which is a regulatory requirement)	3	2	2	12	MAJOR	N	Introduce an active air monitoring location and surface monitoring at the stopper bowl.

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(Table 10.0-1 Continued)

Risk Item	Problem Statement / Requirement	Risk Description / Potential Failure Mode and Impact	Causes of process/ product failure	Current Controls and Preventive Actions/ Comments SEV: assess the risk and impact on sterility and regulatory compliance, in case of a deviation/ deficiency to this statement/ requirement OCC: assess the probability, that the related locations and areas are - in fact- microbiologically contaminated DET: assess the probability, that the implemented EM sampling plan/ frequency/ number of samples/ methods would detect - in fact- a potential contamination	SEV	OCC	DET	RPN	Risk Class	Risk Accepted	Mitigation Measures/ CAPAs
4	Is the air sampling timing and frequency meeting the current industry practices? • Sampling must be done "in operation". • Set- up activities must be included (will be sampled separately to filling operation) • a minimum of 1 active air monitoring sample per shift • continuous passive monitoring (positions may be alternated)	Sampling is not done in operation and misses impact of operators and manufacturing.	Ingress of particles and microbiological contamination operations, or material entry, HVAC failures.	EM program meets requirement. **SEV: 3** (high impact on product sterility if the required sampling is not done properly) **OCC: 1** (it is unlikely product or critical surfaces become contaminated) **DET: 1** (active air monitoring is done during set-up and linked with interventions)	3	1	1	3	MINOR	Y	No action required
8	Are the selected surface monitoring sampling locations: • close to the open product or to product contact surfaces (< 5 cm) • sterile product contact surfaces (= critical surfaces)	GMP deficiency, and potential contamination may not be detected and may result in product contamination within Grade A.	Ingress of particles and microbiological contamination into the high-risk areas by aseptic operations or material entry, HVAC failures.	RABS stopper bowl is not monitored. Other locations are monitored, e.g. filling needles. **SEV: 3** (high impact on product sterility if indirect product contact surfaces = stopper bowl is contaminated) **OCC: 2** (there are several interventions performed, therefore there is a moderate risk) **DET: 2** (no surface monitoring of critical surfaces, which is a regulatory requirement)	3	2	2	12	MAJOR	N	Include additional surface monitoring locations, e.g. stopper bowl

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(Table 10.0-1 Continued)

Risk Item	Problem Statement / Requirement	Risk Description / Potential Failure Mode and Impact	Causes of process/ product failure	Current Controls and Preventive Actions/ Comments SEV: assess the risk and impact on sterility and regulatory compliance, in case of a deviation/ deficiency to this statement/ requirement OCC: assess the probability, that the related locations and areas are - in fact- microbiologically contaminated DET: assess the probability, that the implemented EM sampling plan/ frequency/ number of samples/ methods would detect - in fact- a potential contamination	SEV	OCC	DET	RPN	Risk Class	Risk Accepted	Mitigation Measures/ CAPAs
16	Is the glove monitoring done correctly to recover potential contamination on the fingertip (requires no overlap and a defined minimum contact time)	EM samples of gloves are leading to false negative results and important information will be missed	Important/risky locations are missed due to a wrong location or timing, e.g. RABS gloves fingers close to open product.	Previously, small Rodac plates have been used; based on recommendation, settle plates are now used. Minimum contact time has been defined, no overlap of fingers. **SEV: 3** (high impact on product sterility if the glove sampling is not done properly; contamination may be missed) **OCC: 1** (it is unlikely that the operator gloves are contaminated because good disinfection practices are followed) **DET: 1** (glove monitoring is done after the set-up and as exit monitoring)	3	1	1	3	MINOR	Y	No action required
18	Growth promotion must be supported by batchwise growth promotion testing (GPT). Settle plates should not be desiccated after incubation. Transfer of settle / Rodacs plates into Grade A/B should be done by a one- or two-bag transfer method. Disinfectant spraying on the settle plates should not be done, as this may negatively impact the microbial recovery.	EM samples lead to false negative results and important information will be missed.	Desiccated/ cracked plates, or disinfectant residues prevent recovery of in-house strains.	Supplier has been changed, fewer cracks in the medium now at the site. EM isolates included now in the GPT. Introduction of plates into Grade A RABS will be changed as per training video, within one bag. **SEV: 3** (high impact on product sterility or Grade A if the media plates do not support microbial growth; contamination may be missed) **OCC: 1** (it is unlikely that surfaces within Grade A are contaminated) **DET: 1** (GPT is done batchwise)	3	1	1	3	MINOR	Y	Add more details in EM SOPs

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Figure 10.0-1 Example: BioPhorum Model for a Typical Filling Line

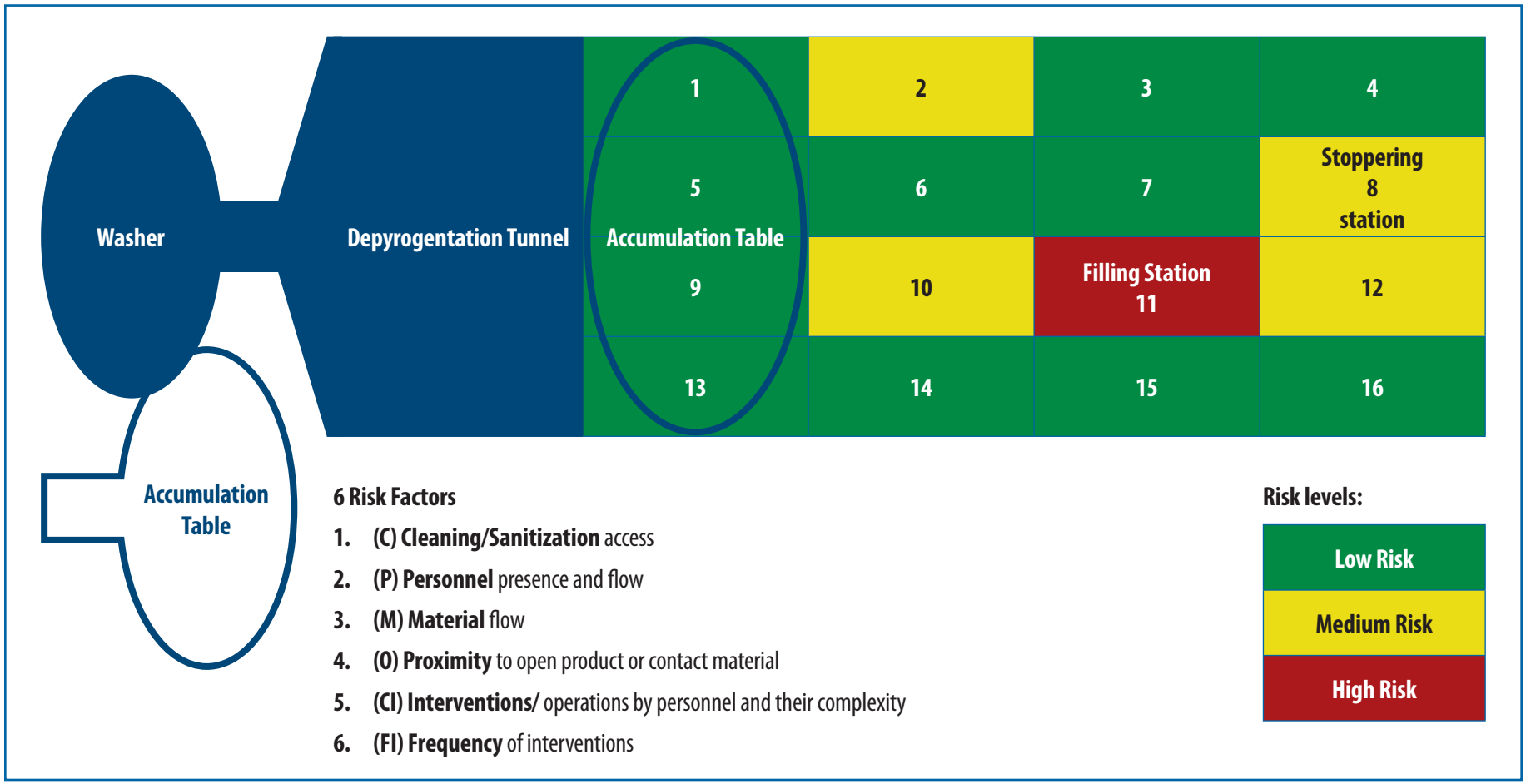

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Table 10.0-2 Filling Line Activities and Associated Risk Rankings for a Closed RABS or isolator filling line

Grid Number	**Process Step Description** (activities in this grid)	**C – Access for Cleaning** (if not automated)	**P – Personnel Presence**	**M – Material Flow**	**O – Proximity to O Product or Exposure to Direct Product Contact Materials pen**	**CI – Complexity of Interventions**	**FI – Frequency of Interventions**	**Risk Score** (x2 weight for Proximity)	**Rank** (Heat Map Coloring)
1,5, 9, 13	Turn table has open vials however no interventions are performed in this grid	1	1	2	8	1	1	16	
2	Within 12 inches of open vials, alpha beta port for material transfer, no interventions with direct product	2	1	2	8	1	1	32	
3,7	Behind dosing vessel within 12-24 inches of open product, no interventions or material transfer	1	1	1	4	1	1	4	
4	Behind stopper station. Some material flow for EM plates.	1	1	2	4	1	1	8	
6	Within 12-24 inches of open vials, behind infeed conveyor, no material transfer or interventions	1	1	2	4	1	1	8	
8, 12	Stoppering station and outfeed star wheel. Material flow for EM plates. Clearing stopper jams with poker less than 1 batch.	1	1	2	8	2	1	32	
10	Part of the needle docking station, needle holder, and stoppering station. Material flow for EM. Clearing stopper jams with sterilized tool occurs less than 1 per batch.	1	1	2	8	2	1	32	
11	Conveyor belt with harder to clean areas. No personnel interventions. Intervention conducted thru glove ports. Material introduced via alpha/beta ports, < 12 inches from open product. Correction of vial jam or line stoppage. Interventions performed with sterile equipment more than 1 time during a batch.	4	1	2	8	2	4	512	
14	In front of starwheel and conveyor. Material flow for EM plates.	1	1	2	4	1	1	8	
15	In front of needle holder and docking station. Material flow for EM plates. No interventions	1	1	1	4	1	1	4	
16	In front of stoppering station. Within 12-24 inches of product. Material flow for EM plate.	1	1	2	4	1	1	8	

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Table 10.0-3 EM Table Template example

Results of air flow studies	Historical data (incl. data on total particulates)	Recent changes which may be relevant	Notes	Recommended Monitoring					ZONE DECISION
				During Set-Up		During process (2 times during each batch -beginning and end)		End of Process (Surface)	
				Total particulates	Viables	Total particulates	Viables	Viables	

10.1 Example of an HACCP

Hazard analysis and critical control points (HACCP) is a formal qualitative risk assessment technique that is designed to identify critical steps in a process that require control and monitoring to ensure that hazards are not realized, with the goal of developing a comprehensive monitoring plan. For the purposes of this example, the types of hazards analyzed are biological (i.e., viable monitoring) and physical (i.e., total airborne particulates).

Figure 10.1-1 outlines the seven principles of HACCP as well as how the principles can be applied to develop an EM program. **Table 10.1-1** provides a practical example of an HACCP exercise applied to a Grade C cell culture room with an occupancy of twelve operators.

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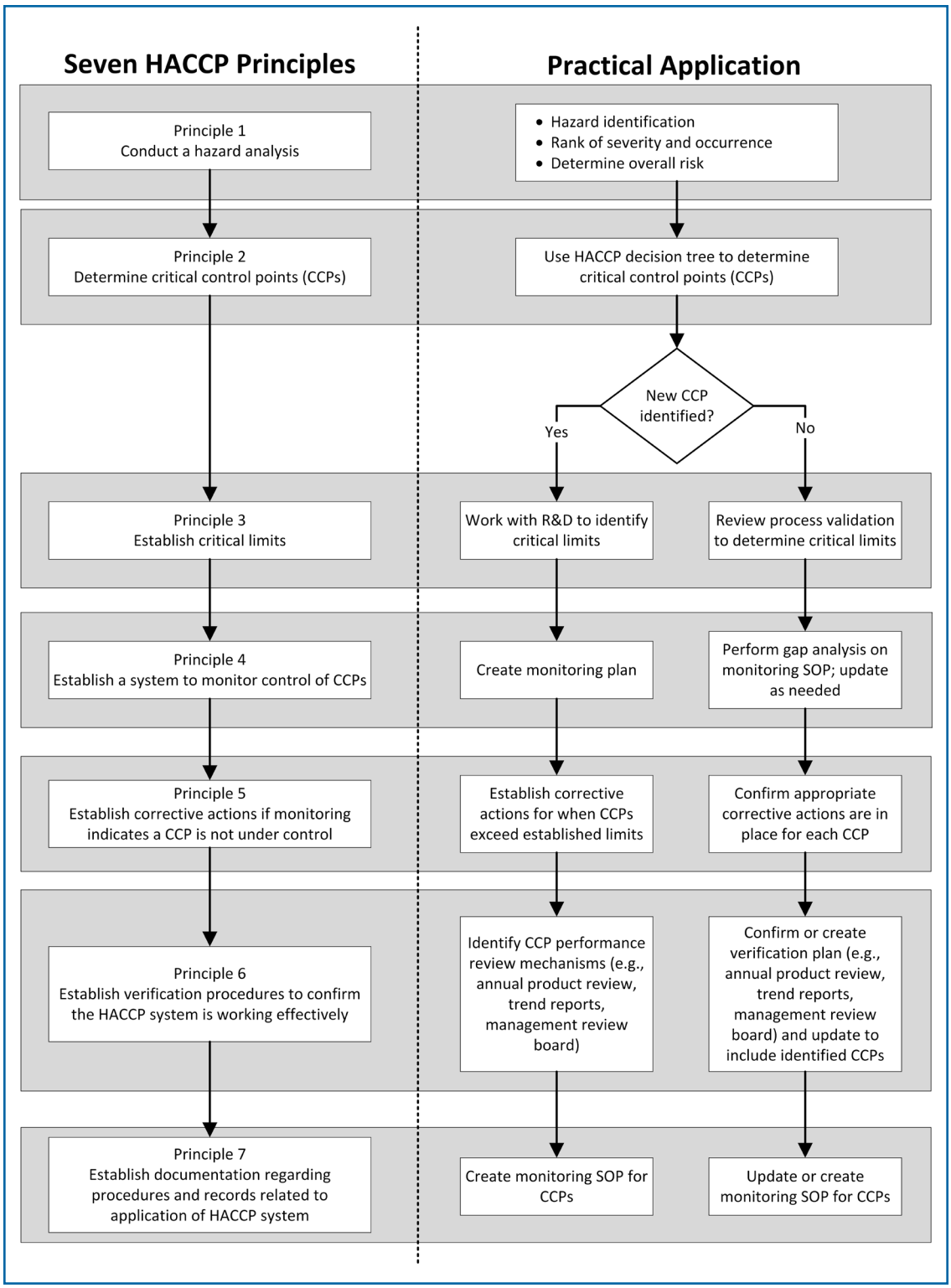

Figure 10.1-1 HACCP Principles and Application

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Table 10.1-1 Application of HACCP Methodology to Pharmaceuticals (from WHO Technical Report Series, No. 908, 2003, Annex 7)

	Hazard Identification Controls Evaluation			Risk Analysis and risk Evaluation						Critical Control Points Determination				Risk Control
Room Sector	Potential source of microbial contamination	Hazard	Current Controls	Likelihood	Likelihood rationale	Severity	Severity rationale	Overall risk level	Risk Acceptable? (Yes/No)	Q1: Is sampling at this location required for cGMP?	Q2: Is sampling at this location critical to protect patient safety or product quality?	Q3: Will monitoring at this location provide data needed to inform a robust investigation?	Q4: Is this location a Critical Control Point?	**Mitigations**
1	Phone (touch point) Computer workstation (touch point) Bioreactor (×3)	Exceed the EM alert level/ action limit (viable)	Examples of controls to list include: personnel, waste, and material flow SOPs; training on cleanroom behavior; EMPQ results; etc.	Occasional	One action level excursion reported in last two trended periods.	Critical	Impact to Grade B/C area.	High	No	No	Yes	N/A	Yes	CAPA 12245
1		Exceed the particulate alert level/ action limit (nonviable)		Remote	No alert or action level excursions during last two trended periods.	Critical	Impact to Grade B/C area.	Low	Yes	No	Yes	N/A	Yes	N/A
2		Exceed the EM alert level/ action limit (viable)		Unlikely	Two alert level excursions reported in the last two trended periods.	Critical	Impact to Grade B/C area.	Medium	Yes	No	Yes	N/A	Yes	N/A
2		Exceed the particulate alert level/ action limit (nonviable)		Remote	No alert or action level excursions during last two trended periods.	Critical	Impact to Grade B/C area.	Low	Yes	No	Yes	N/A	Yes	N/A
3		Exceed the EM alert level/ action limit (viable)		Unlikely	Two alert level excursions reported in the last two trended periods.	Critical	Impact to Grade B/C area.	Medium	No	No	Yes	N/A	Yes	N/A
3		Exceed the particulate alert level/ action limit (nonviable)		Unlikely	Two alert level excursions reported in the last two trended periods.	Critical	Impact to Grade B/C area.	Medium	No	No	Yes	N/A	Yes	N/A

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Fundamentals of an Environmental Monitoring Program

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